Machine Learning I
MATH60629A

Modern Generative Models
— Week #11

Today's Plan*

- "Predict" an image
- Generative Models
  - Images (P(x)):
    - Frameworks: Variational auto-encoders (VAEs), Generative Adversarial Networks (GANs)
  - Images conditioned on text (P(x | y)) :
    - Dall-E 2, Imagen

* Like last week, the slides are mostly from David Berger

Auto-Encoder (AE)

Auto-encoder - Recall

For an AE with a single hidden layer:

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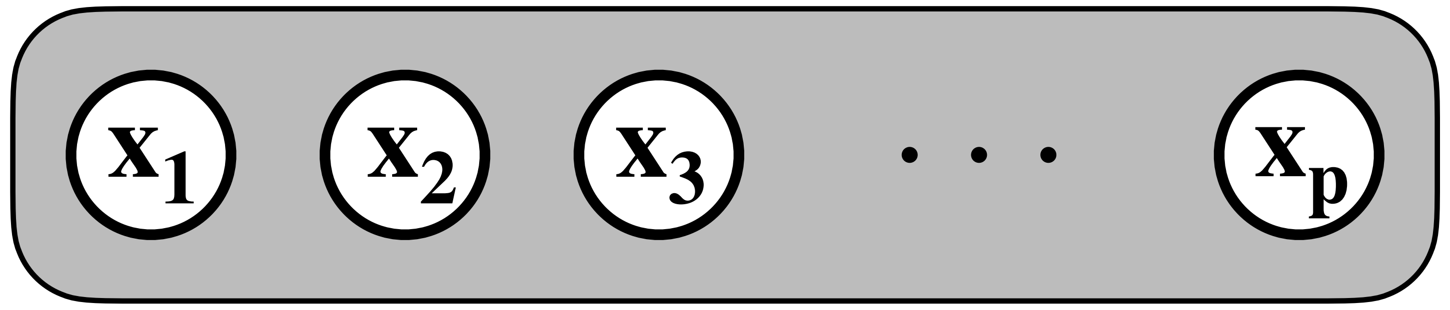
$\mathbf{x}$



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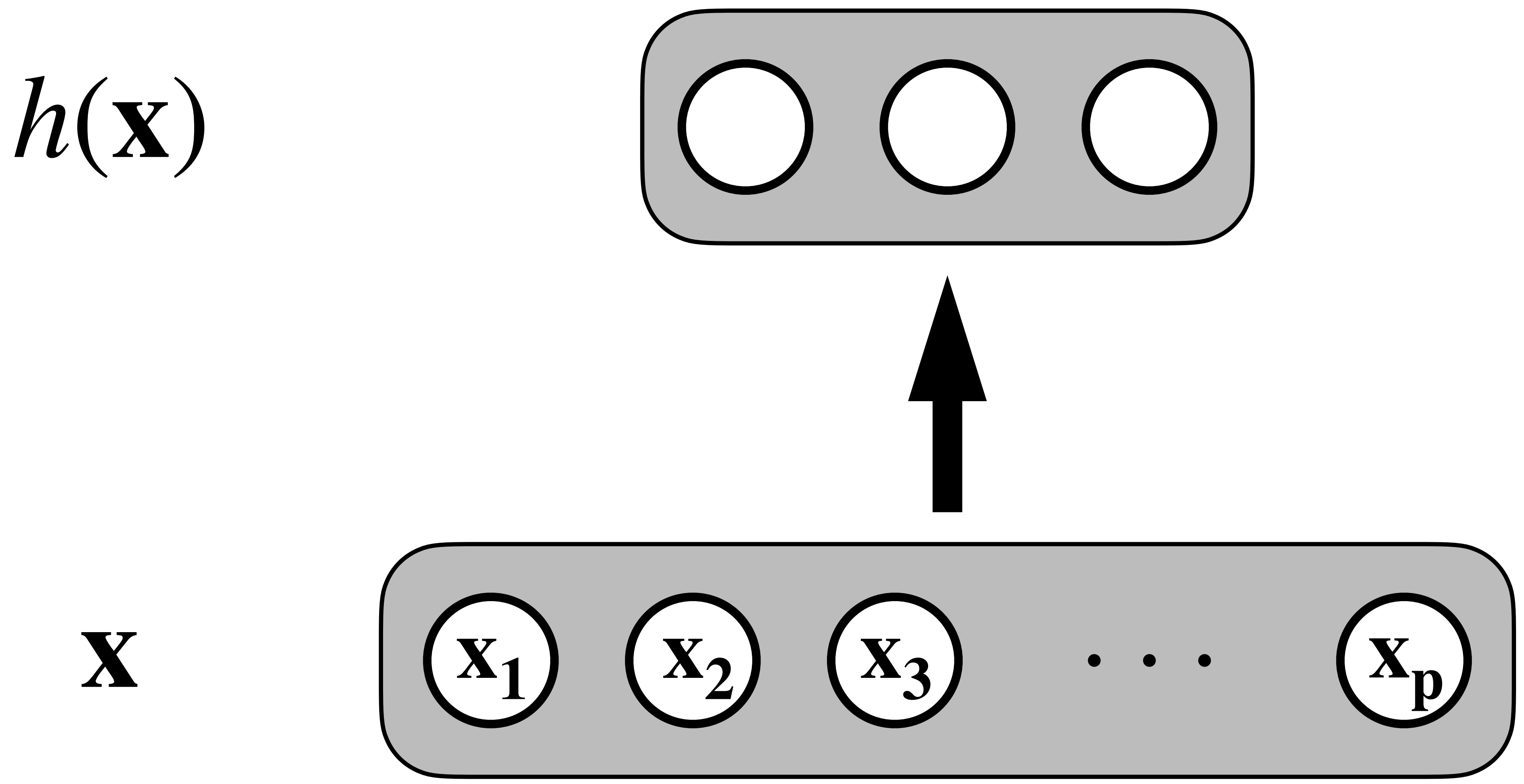

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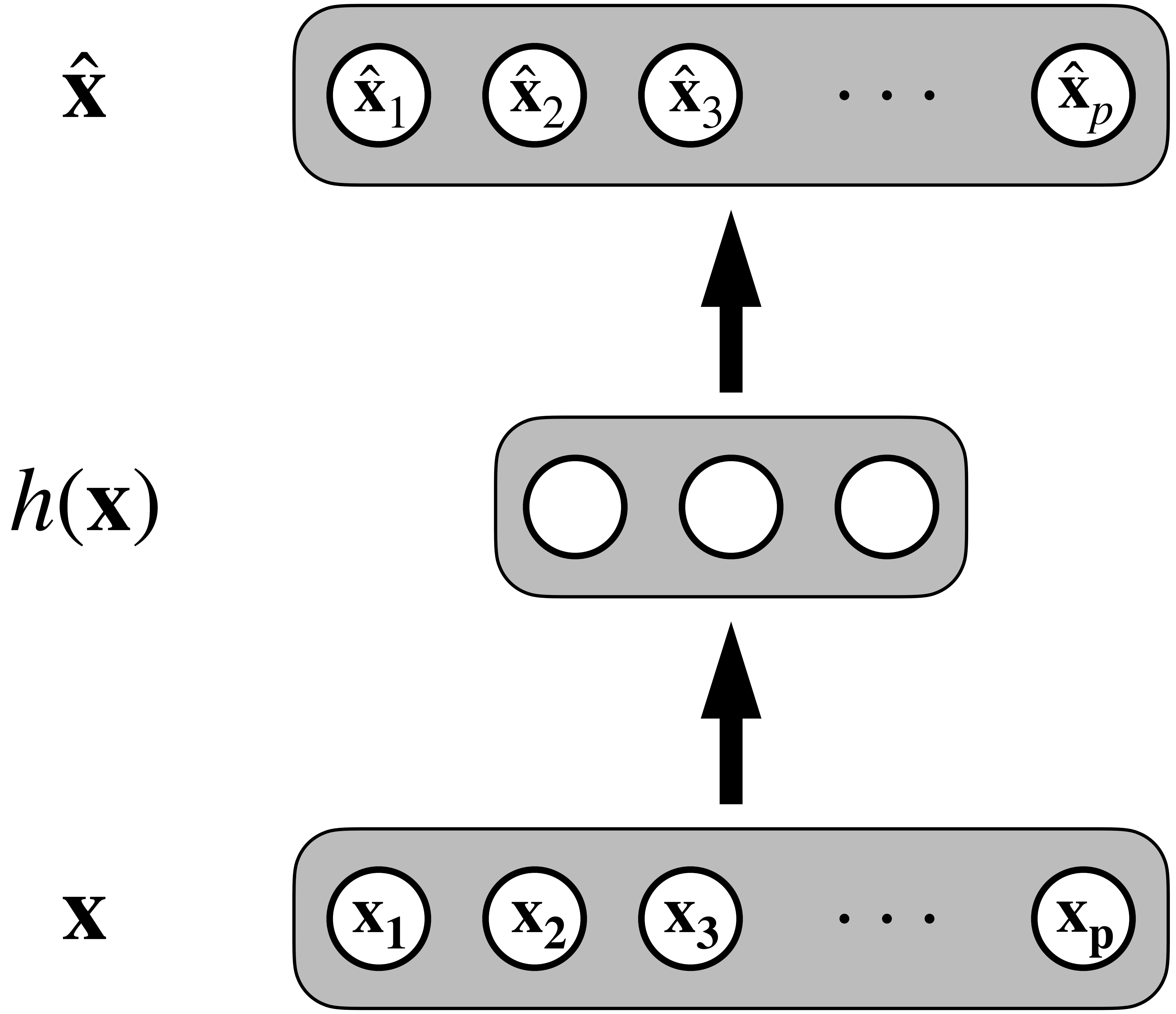

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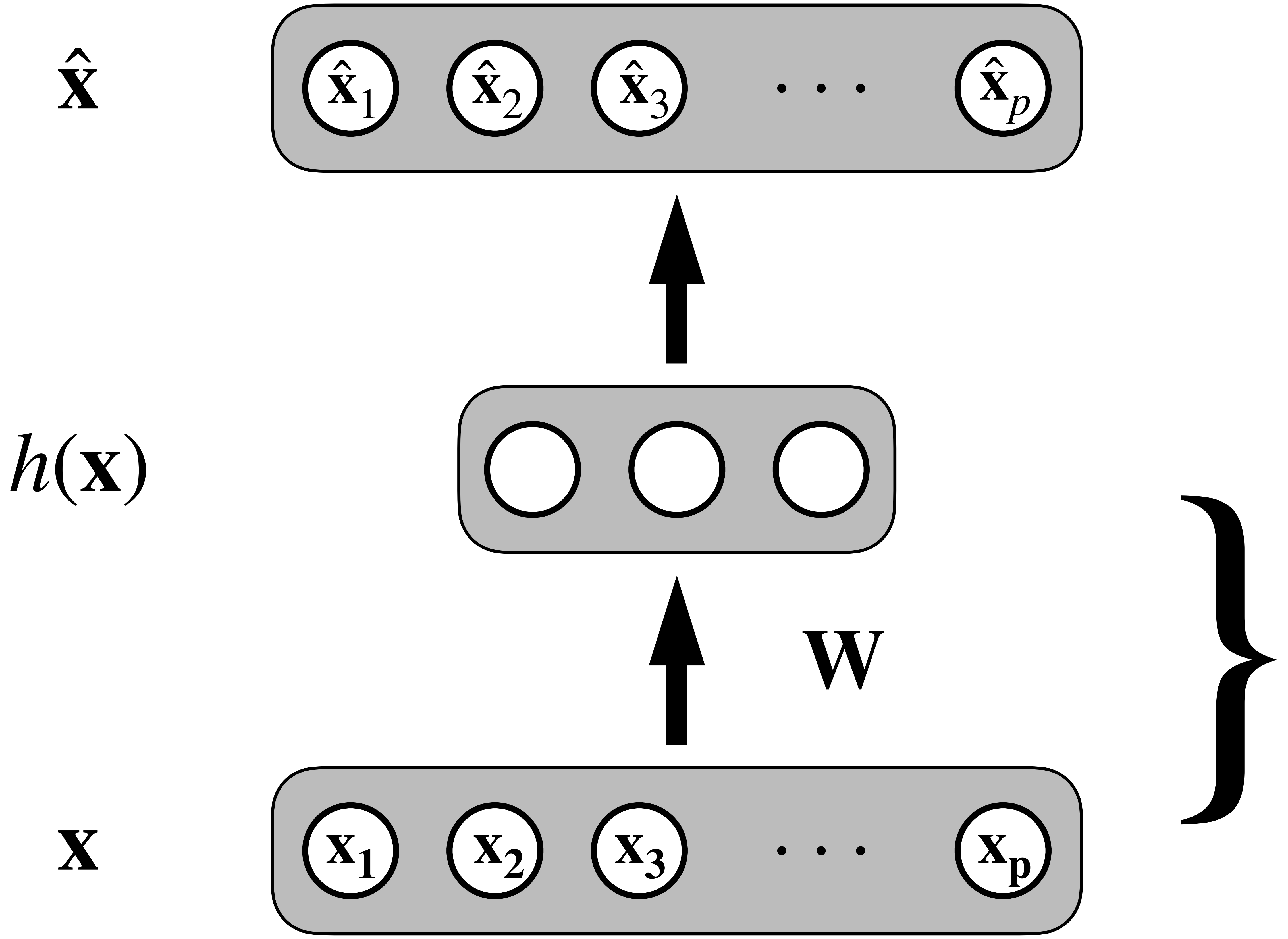

Encoder

$$
\begin{aligned}
h(\mathbf{x}) &= g(\mathbf{a}(\mathbf{x})) \\
&= g(\mathbf{W}\mathbf{x} + \mathbf{b})
\end{aligned}
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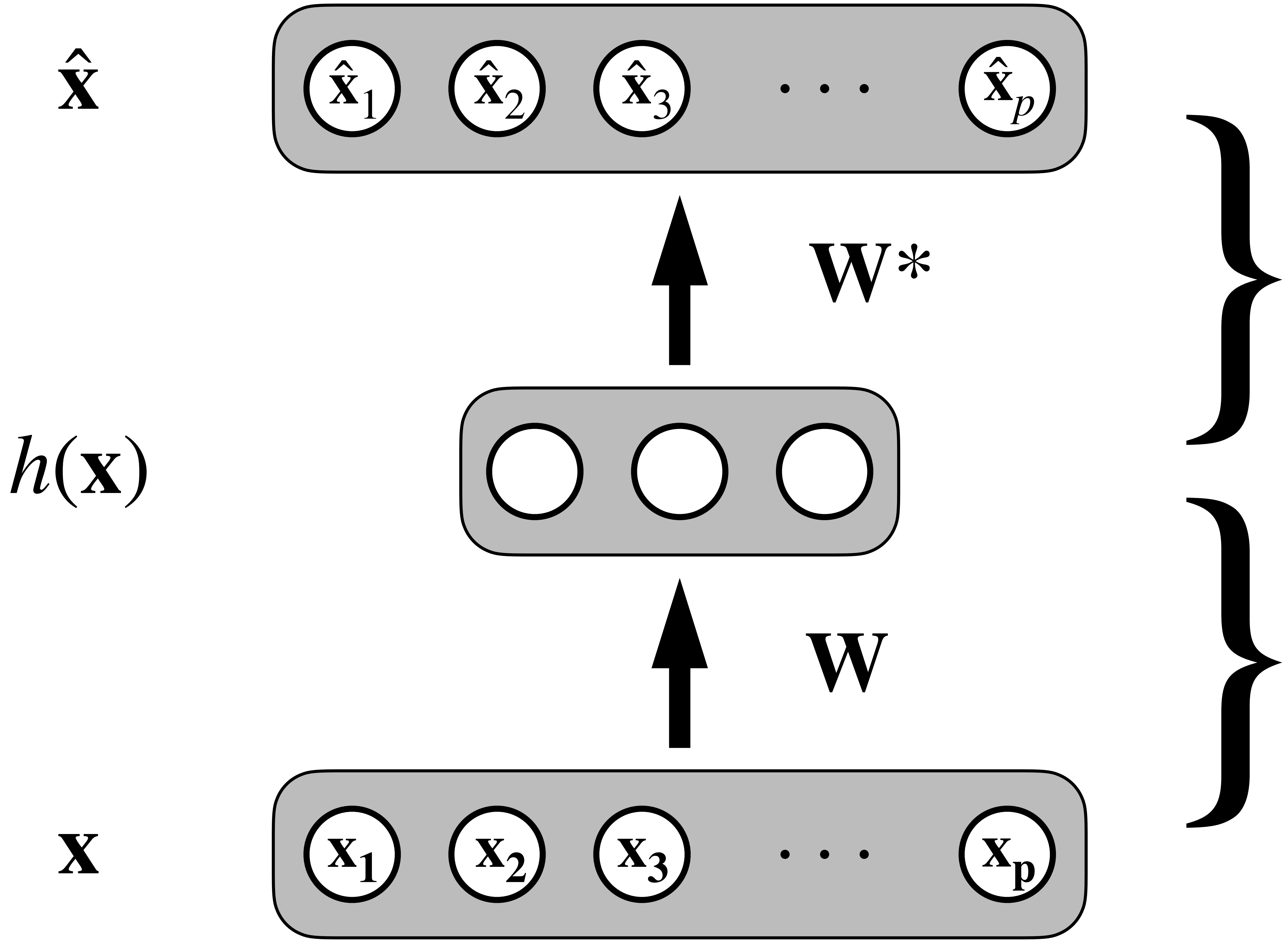

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Variational Auto-encoder (VAE)

* Understanding these models precisely requires concepts that are beyond our class. Here, we aim for familiarization with the terms and an intuitive understanding.

Variational Auto-Encoder - Motivation

Idea:

- The data are generated conditioned on a random variable (Z):

$$\mathbb{P}_{\theta}\left(\mathbf{x} \mid \mathbf{z}\right)$$

You can think of Z has an embedding.
VAEs will learn a prior P(z) and
a posterior P(z|x) over it.

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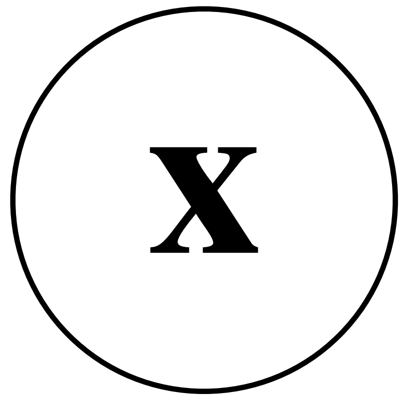

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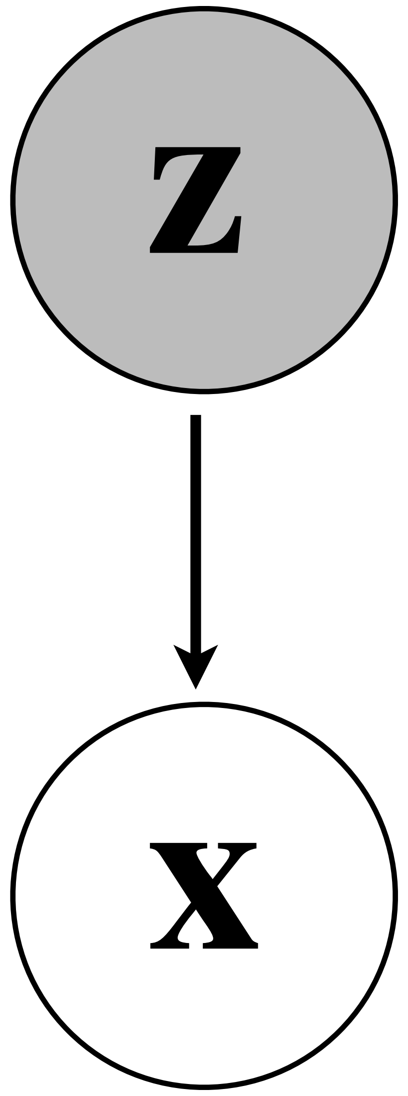

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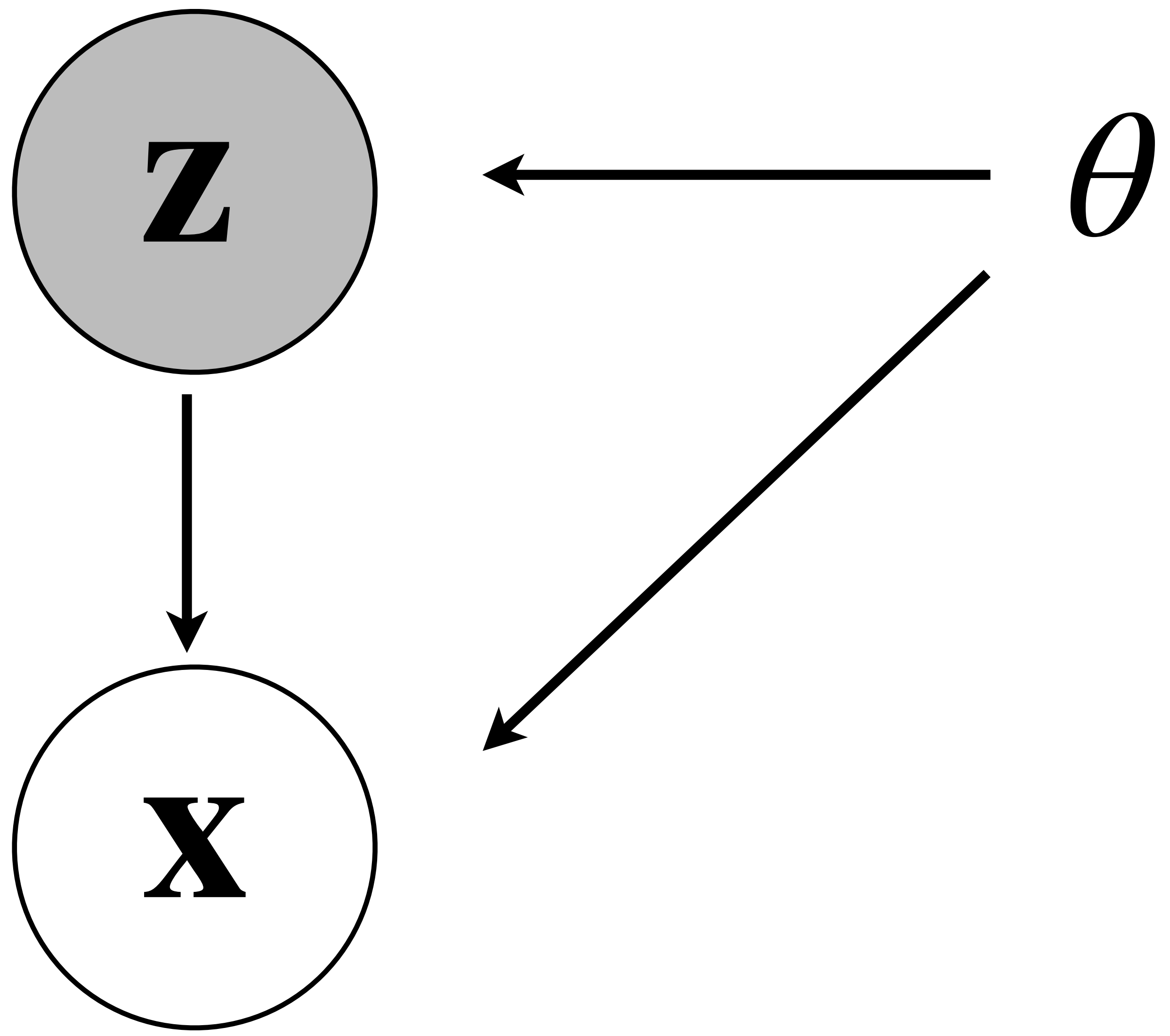

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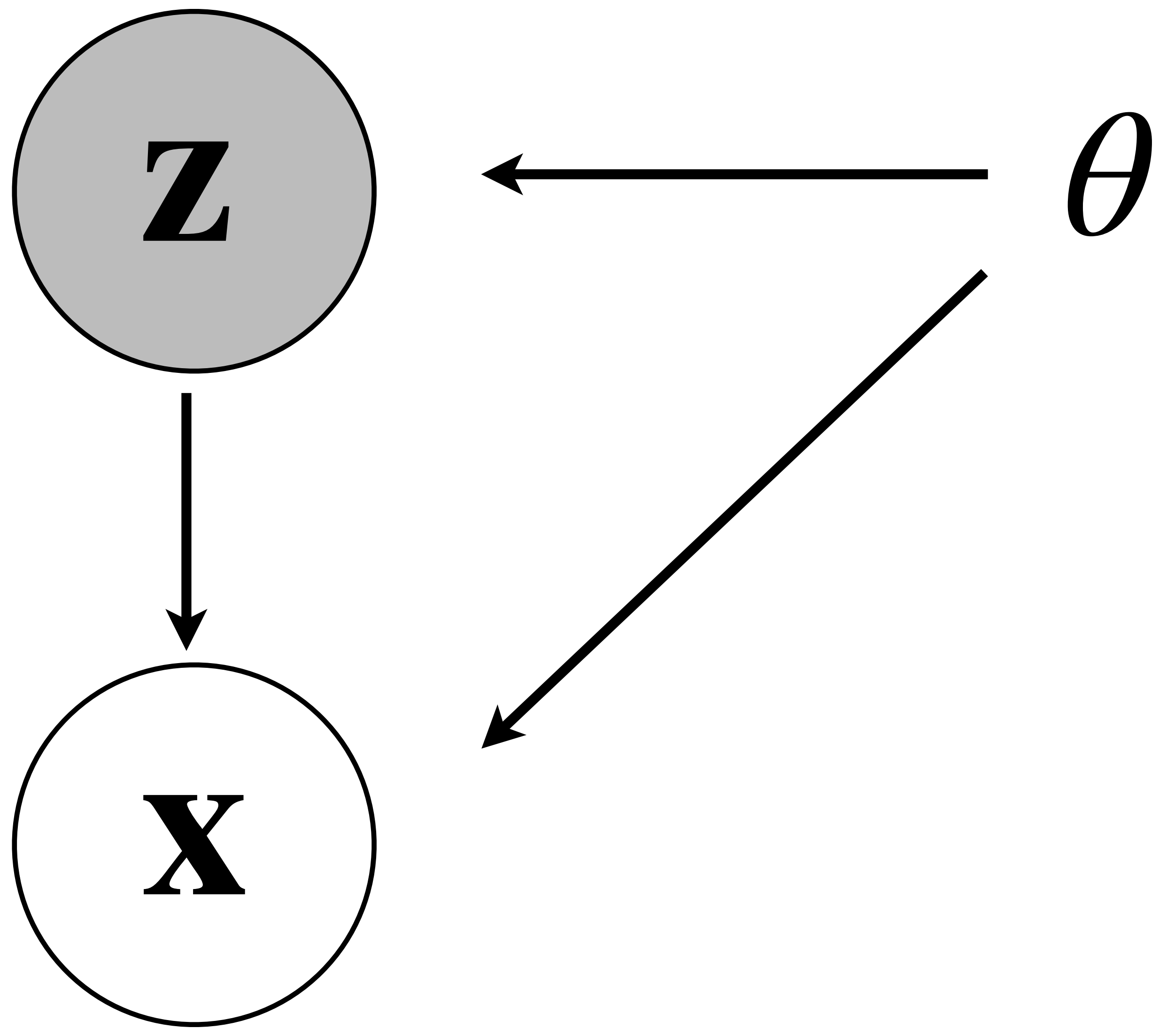

Question:

- How do we learn such a distribution?

Variational Auto-Encoder - Motivation

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Partial answer:

- A "good" model should maximize the likelihood of the data P(x)
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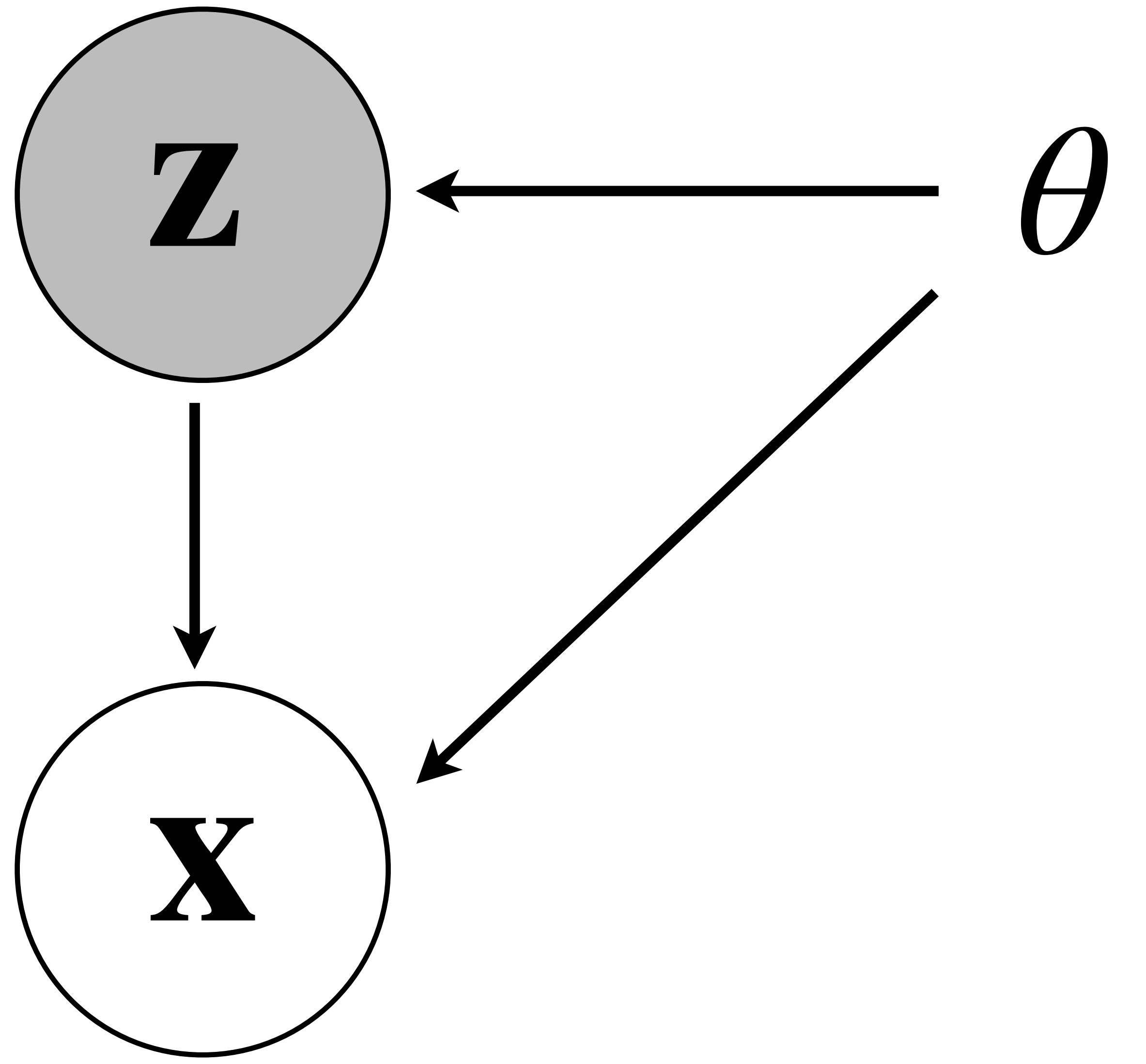

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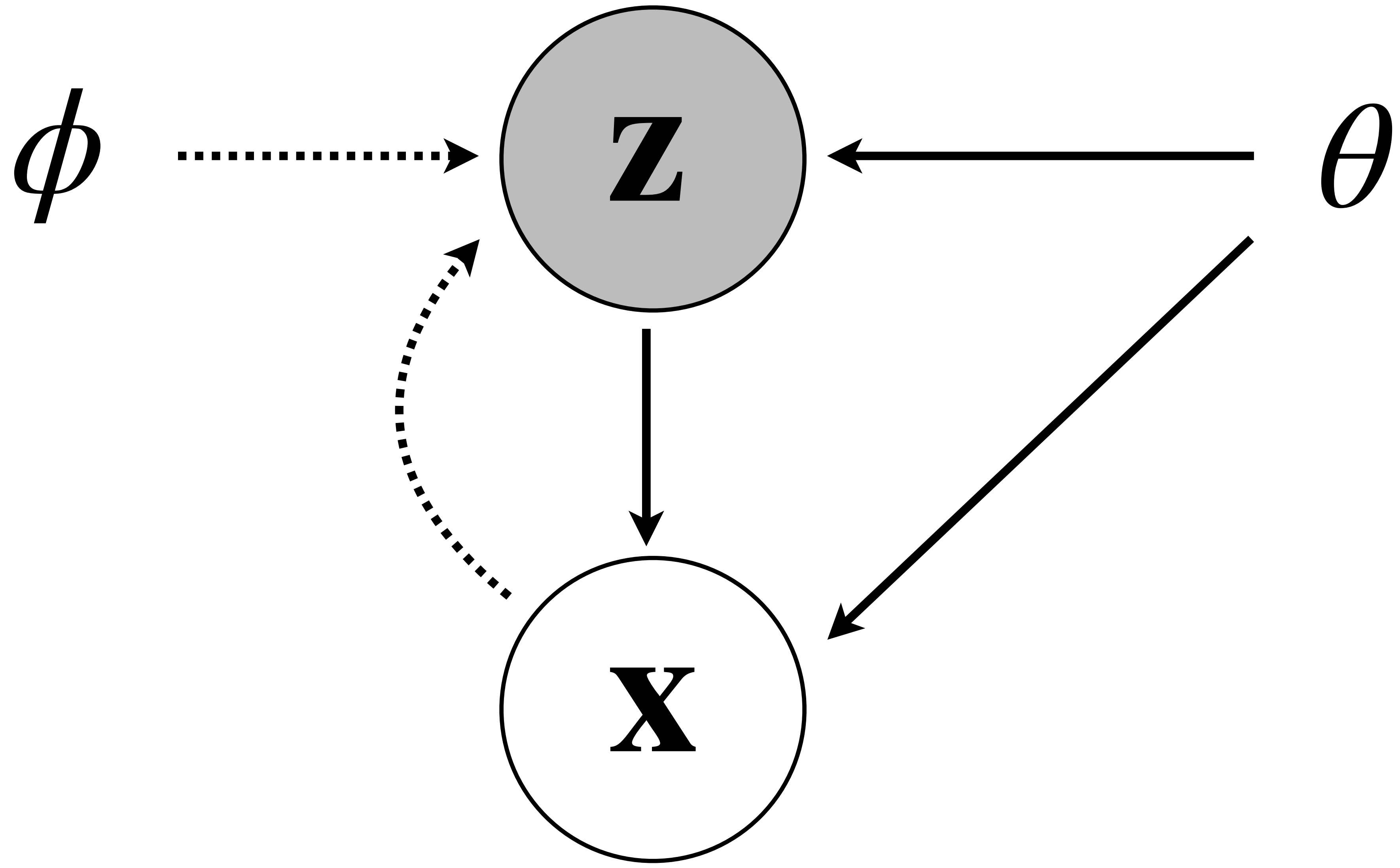

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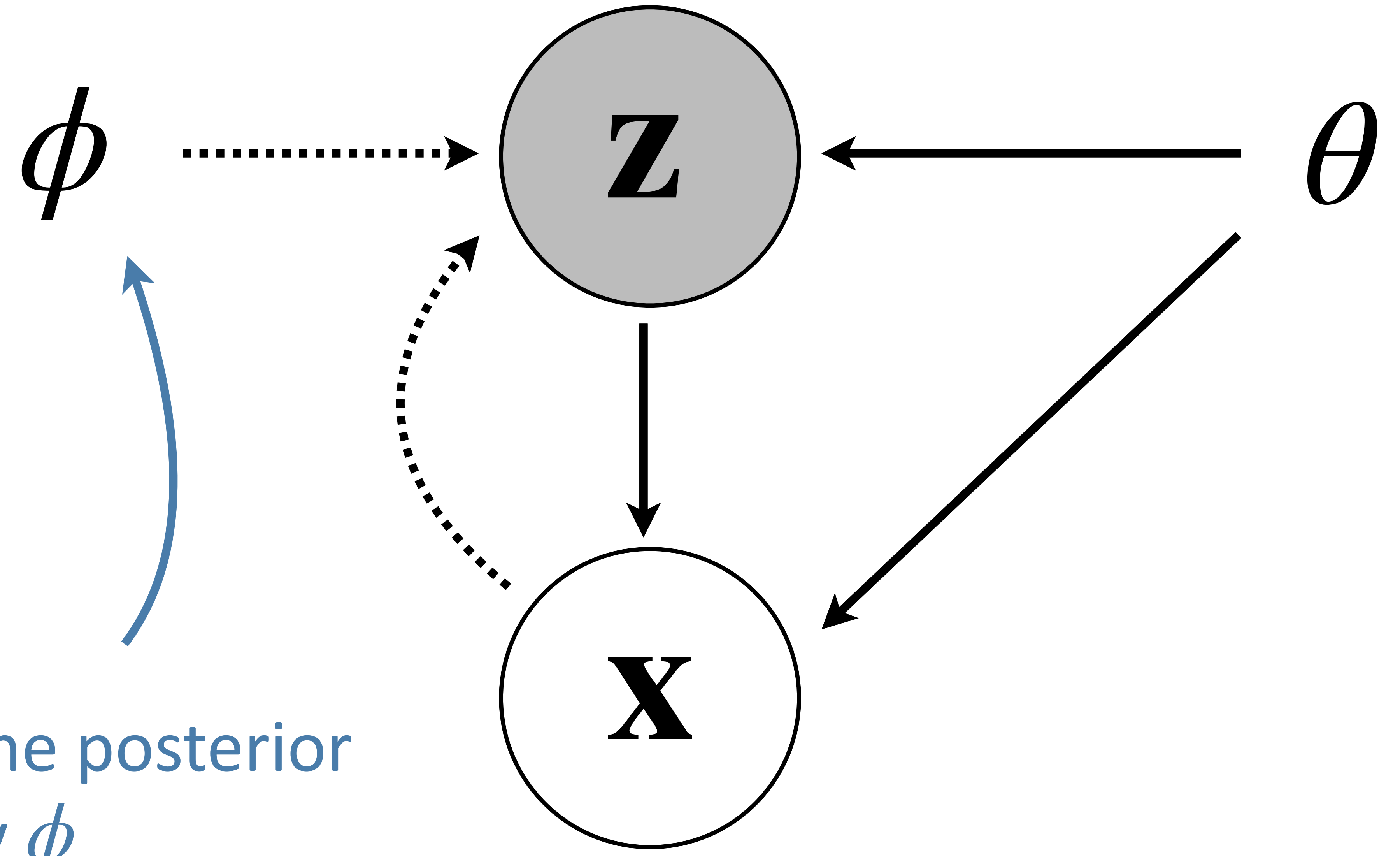

The approximation of the posterior would be parametric by ϕ

Variational Auto-Encoder - Formalism

We can obtain a bound on the log-likelihood of $\mathbf{x}$:

$$\log p_\theta(\mathbf{x}) \;\geq -\,\mathsf{KL}\{q_\phi(\mathbf{z} \mid \mathbf{x}) \;||\; q_\phi(\mathbf{z})\} \;+\; \log p_\theta(\mathbf{x} \mid \mathbf{z}),$$

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- How do we do this in practice?

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Graphical view:

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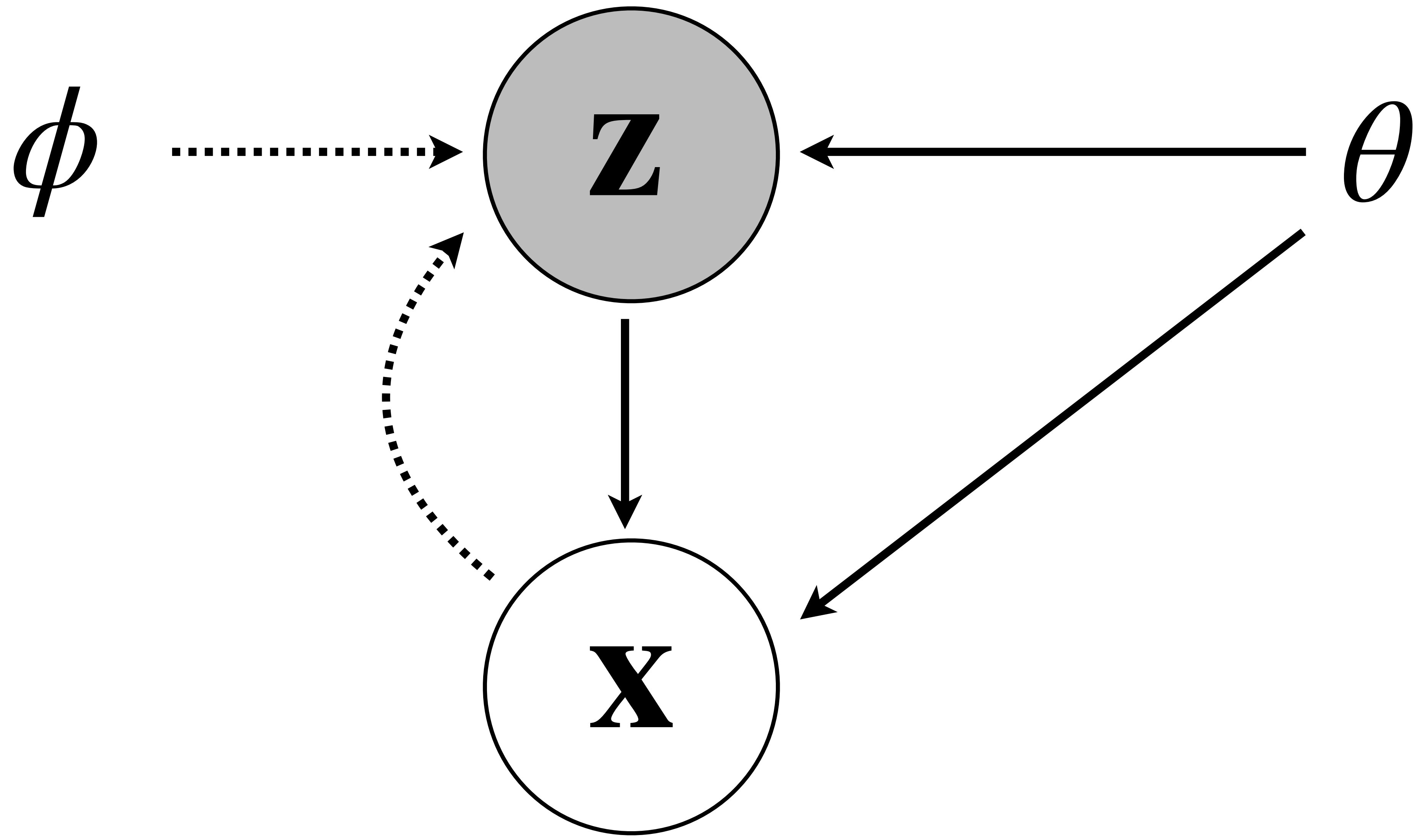

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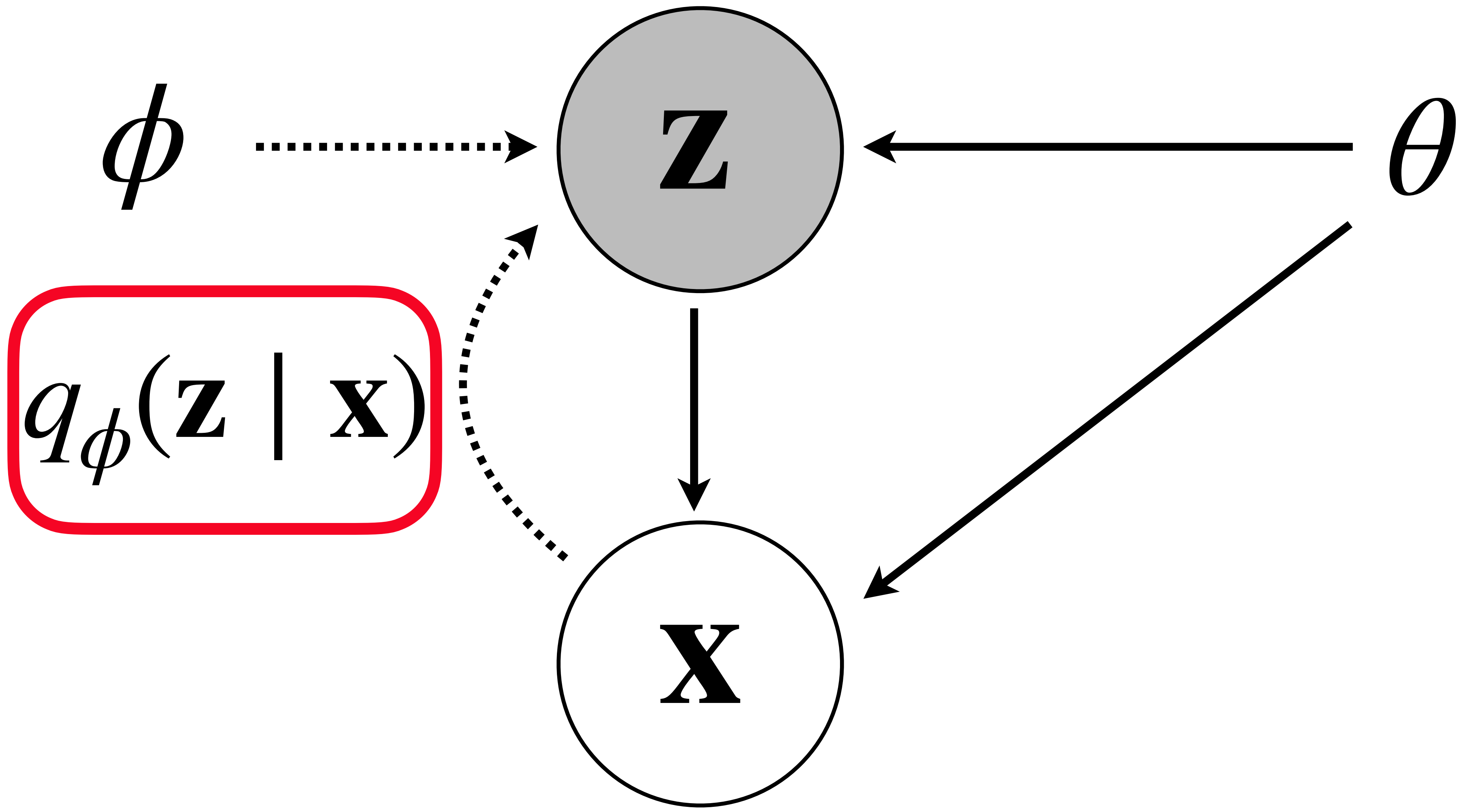

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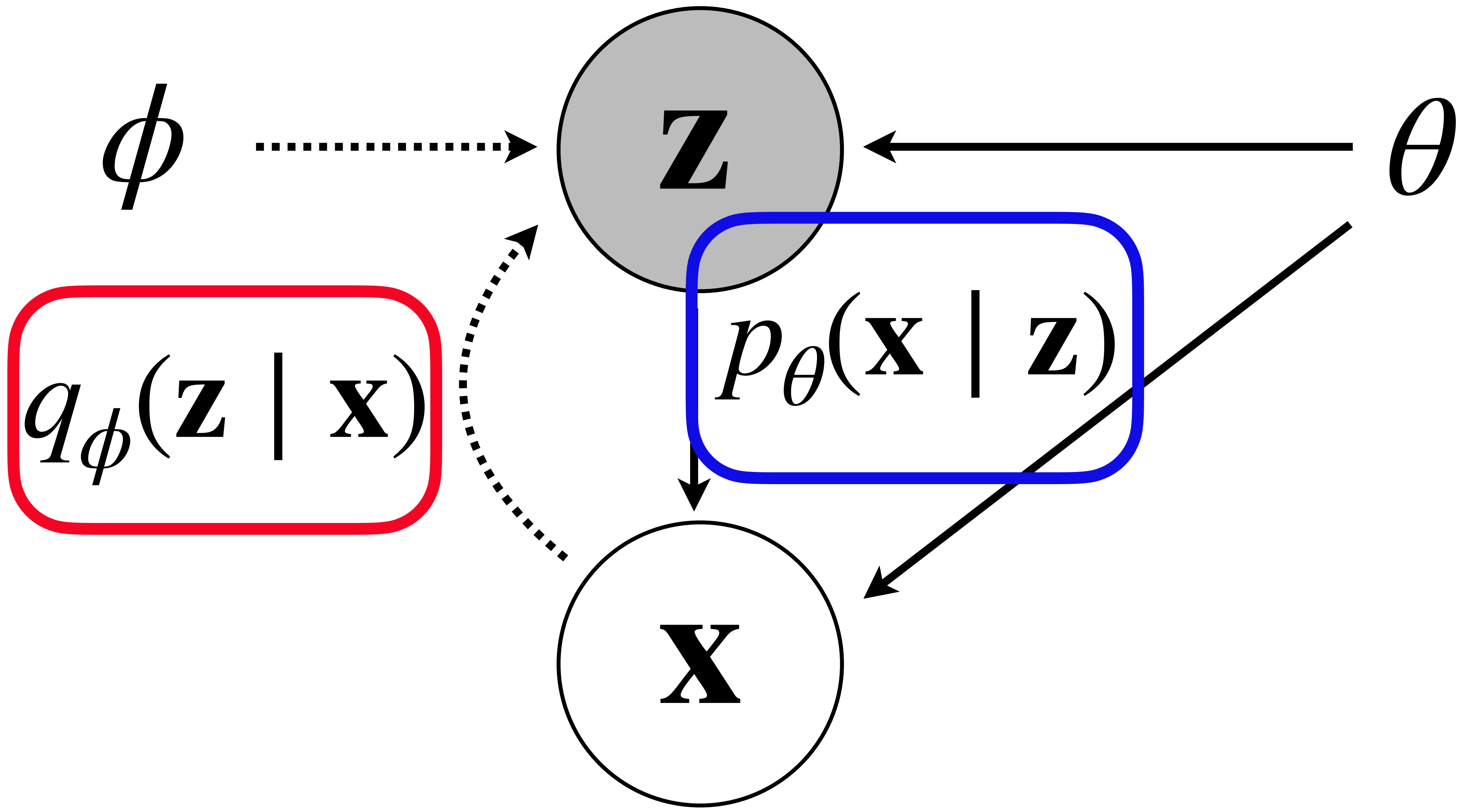

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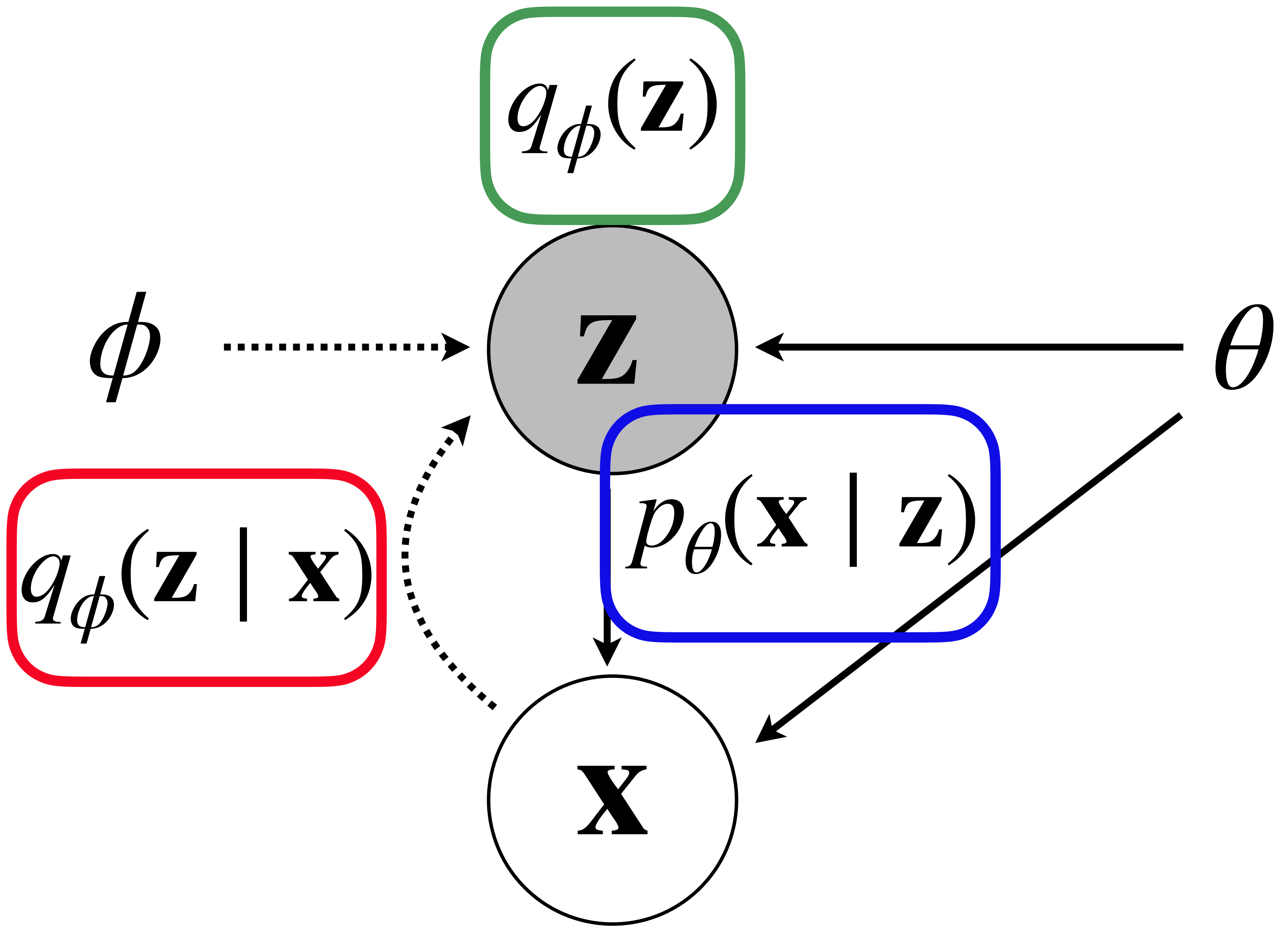

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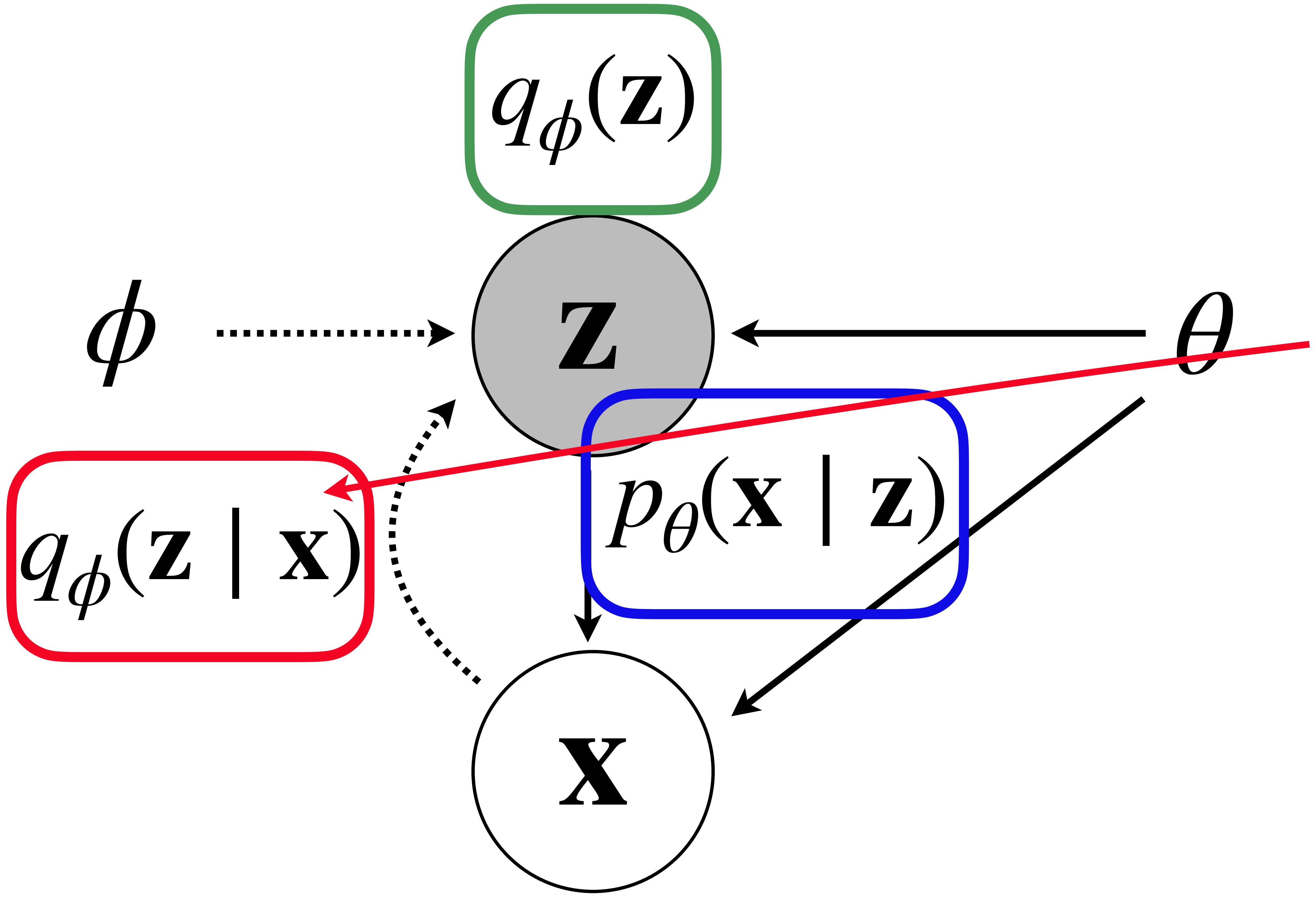

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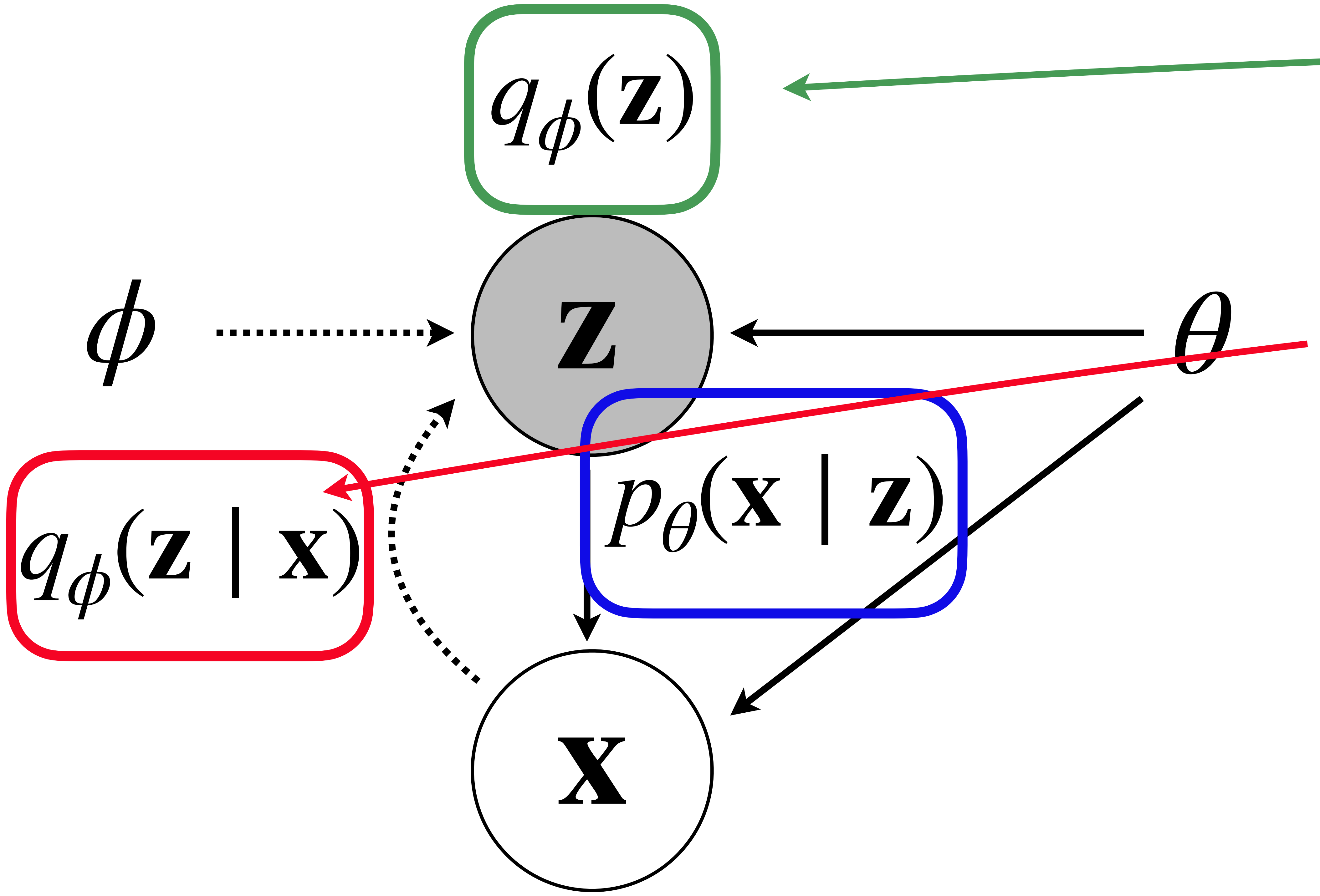

We suppose the prior follows a particular distribution

(e.g., a zero-mean Gaussian)

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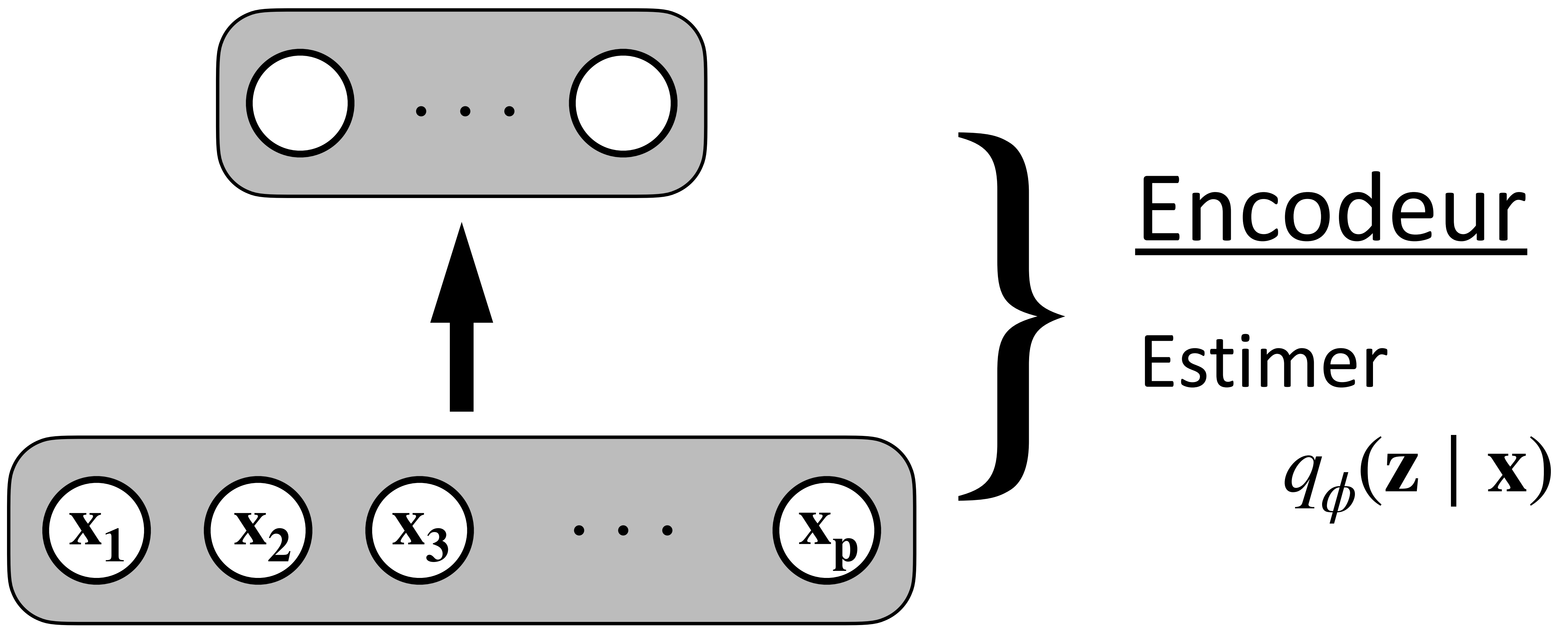

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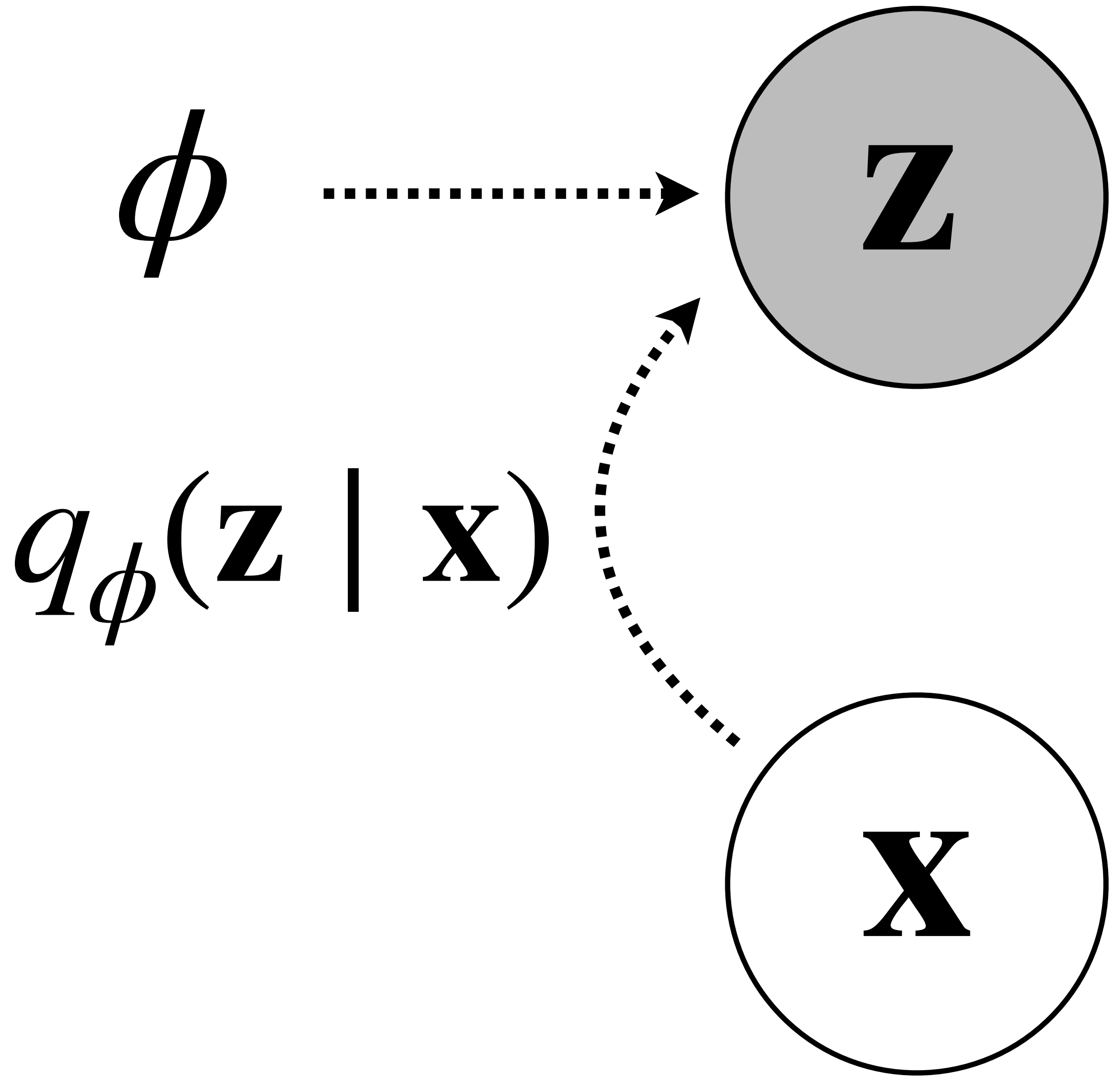

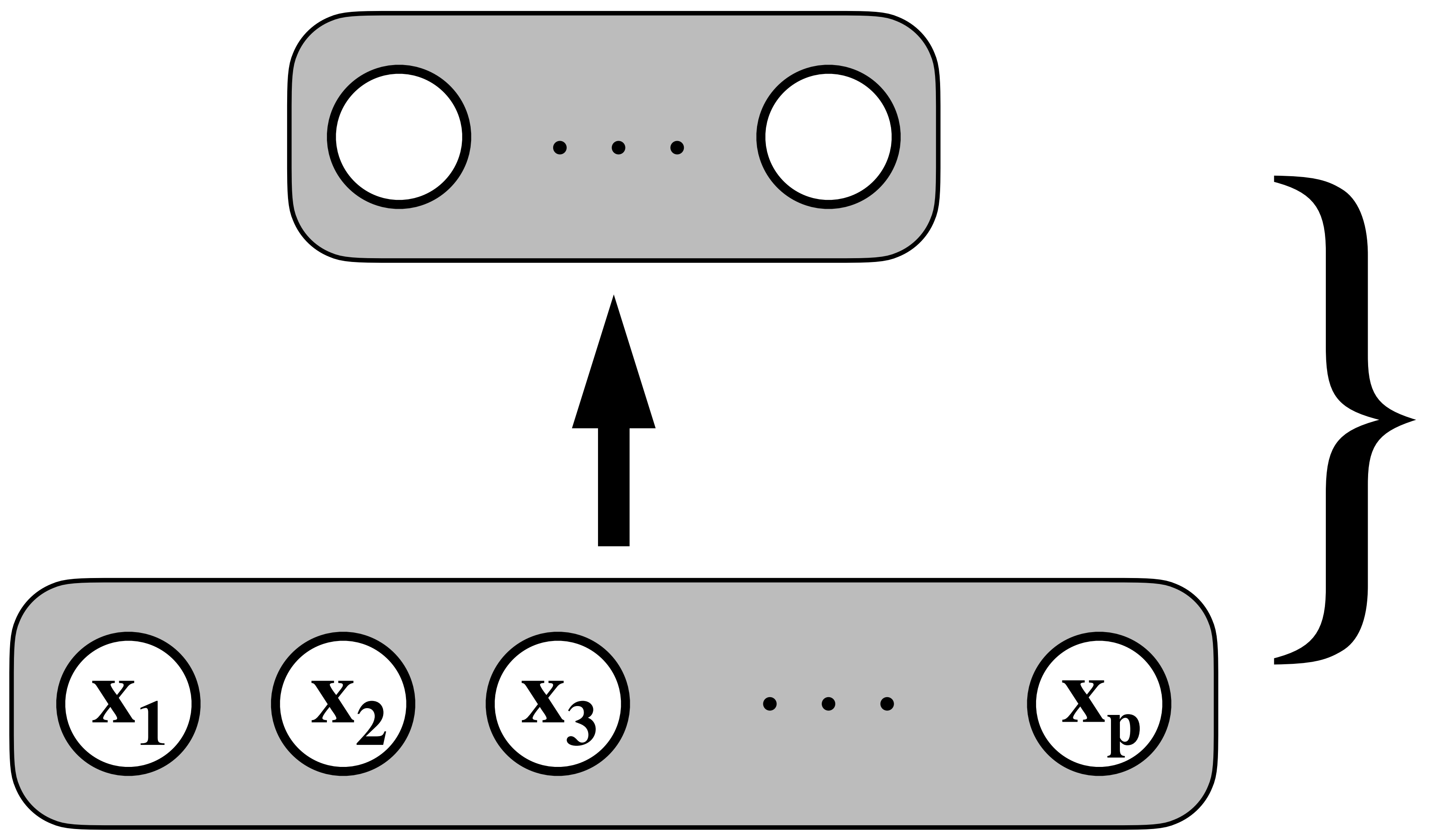

Encodeur

Estimer

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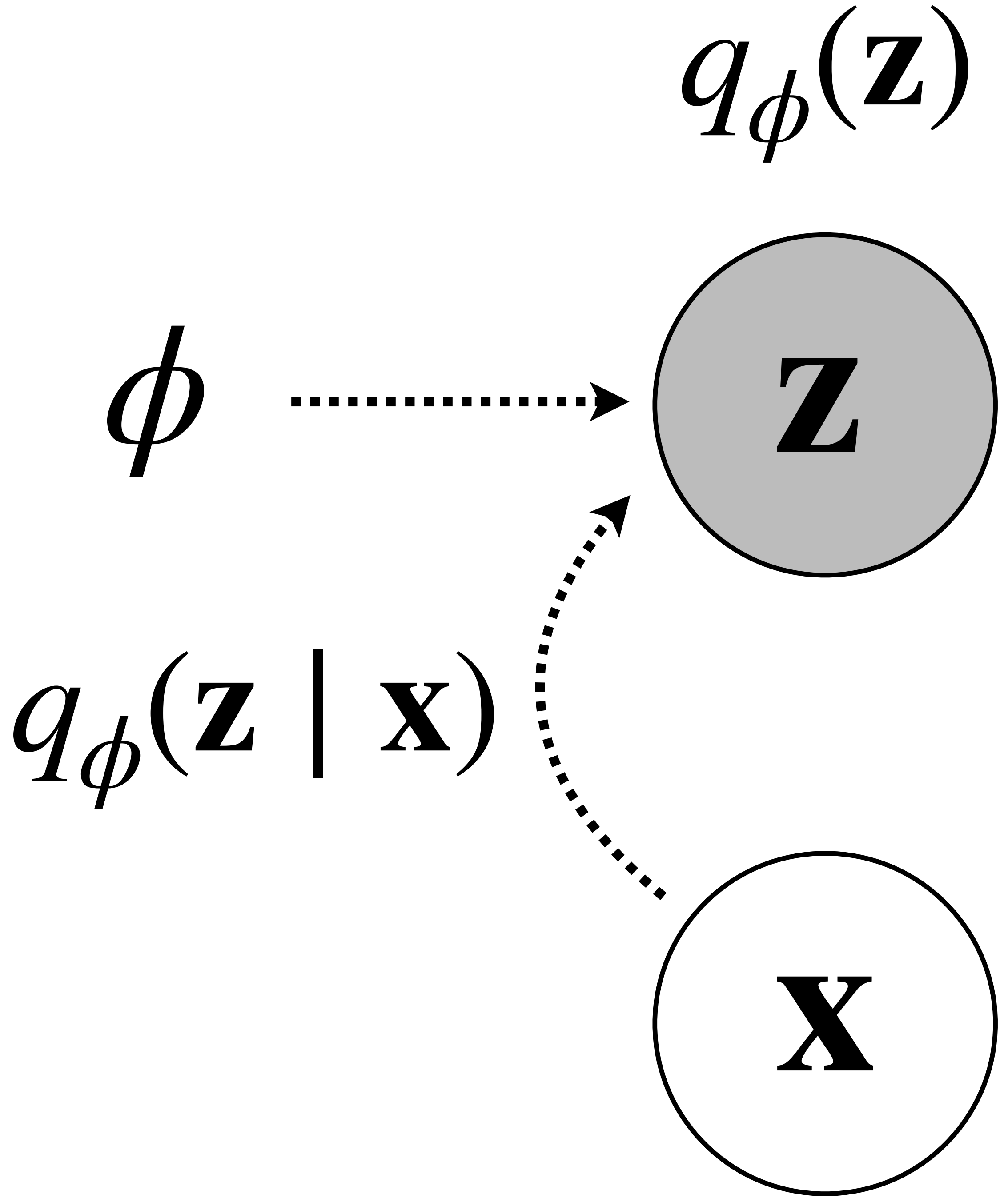

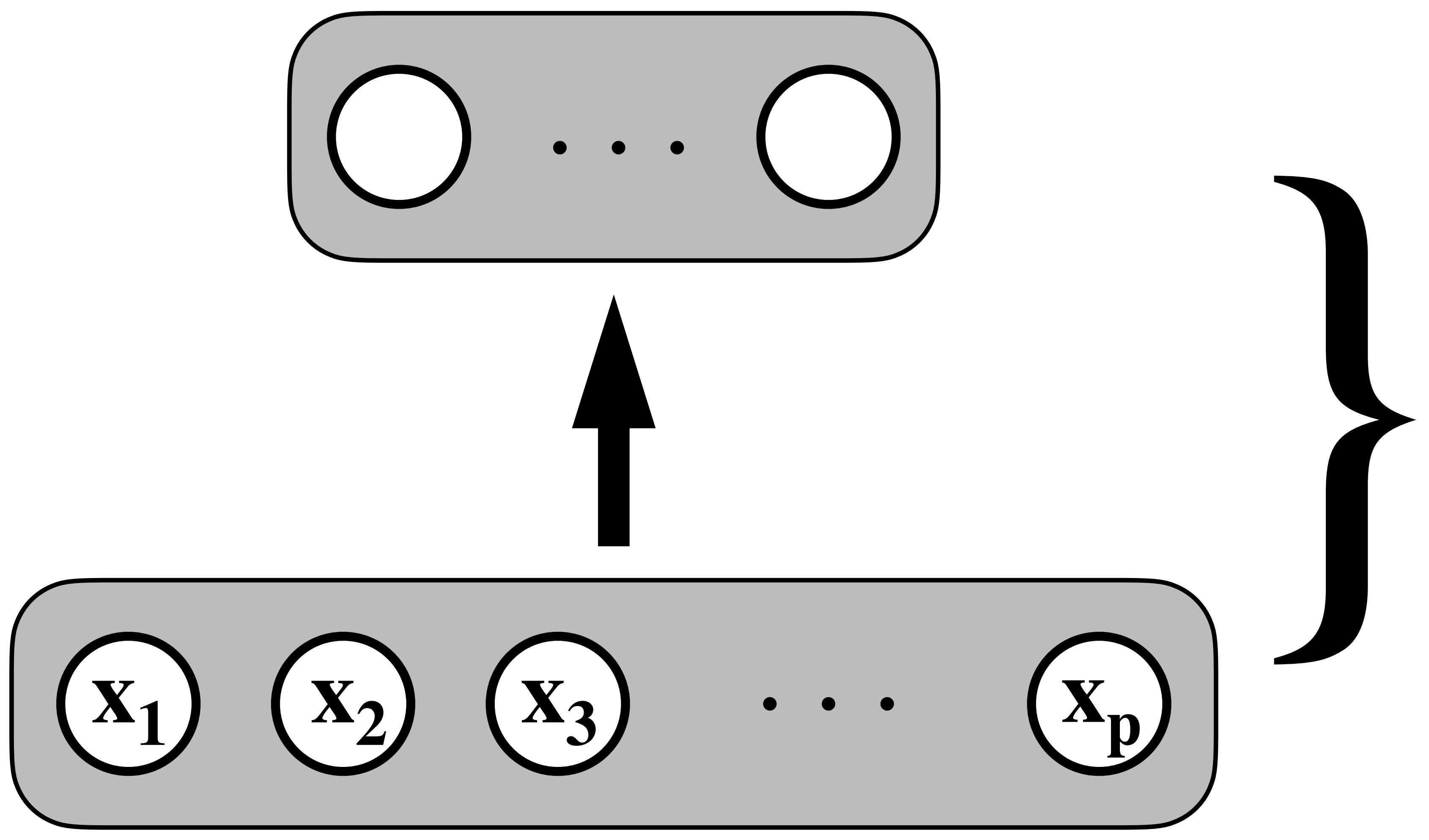

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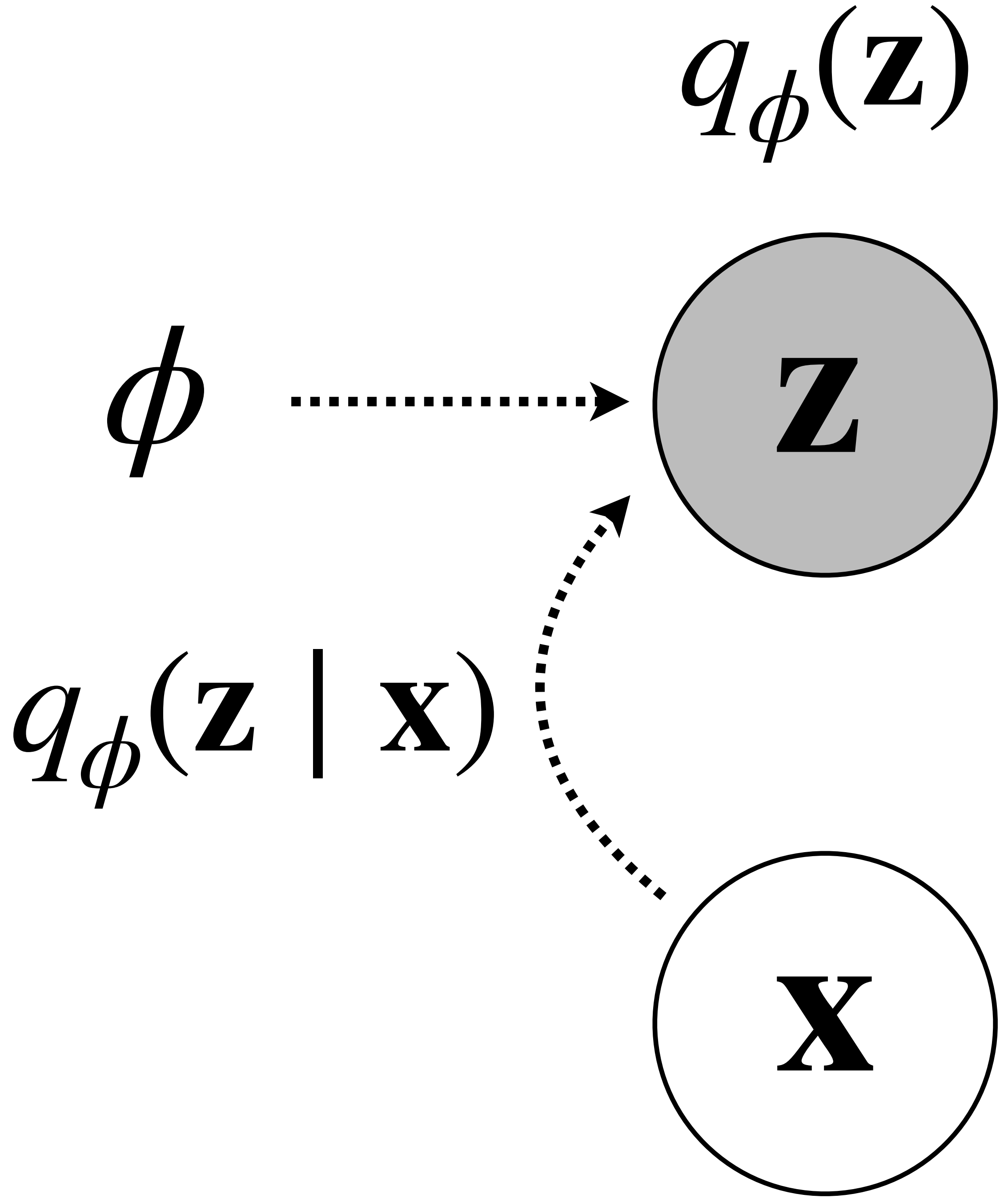

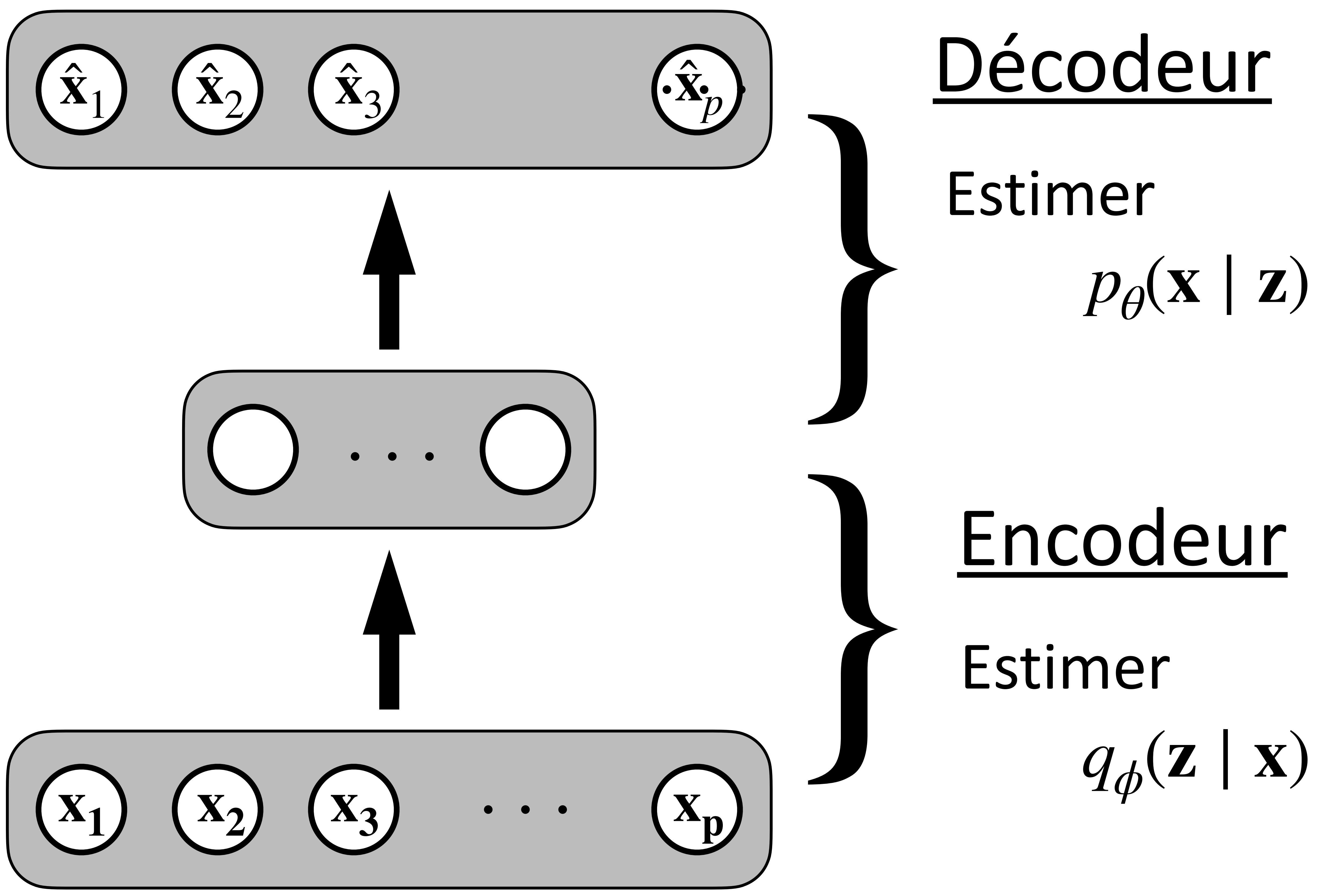

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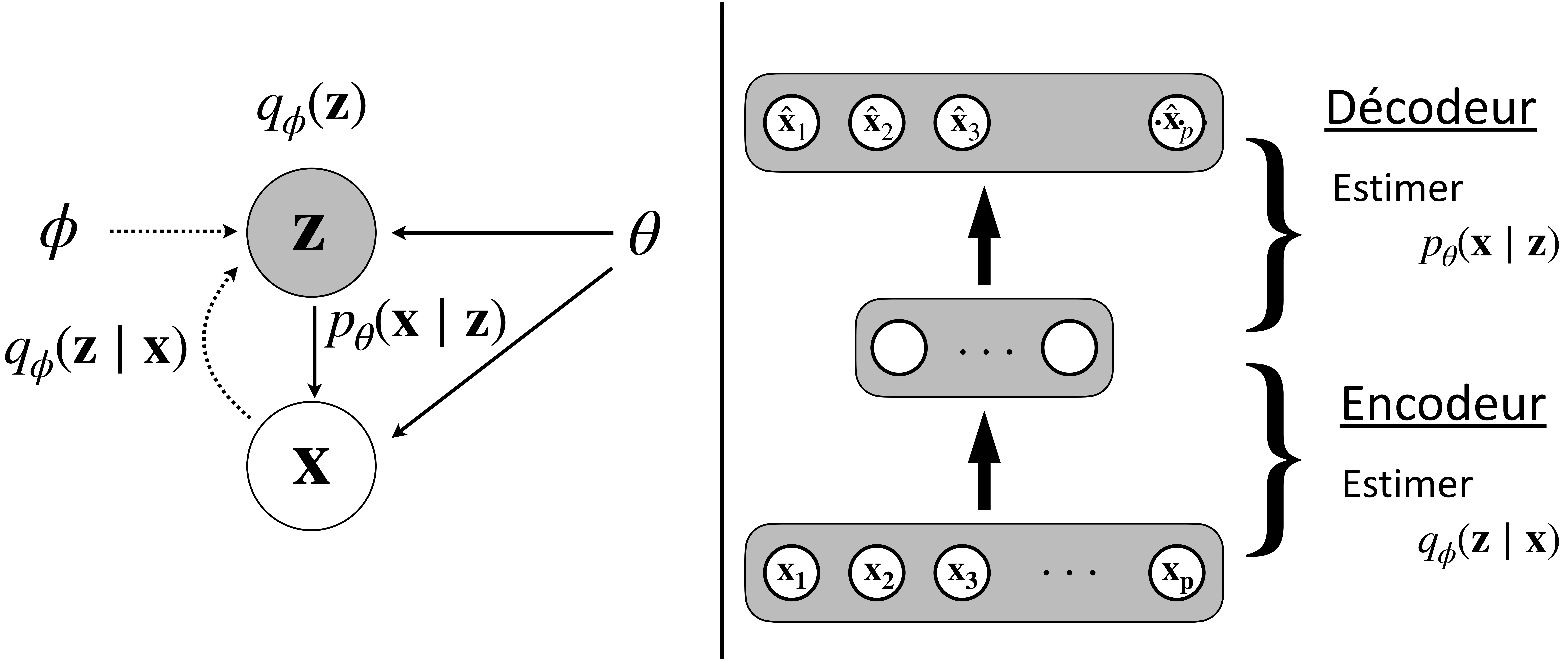

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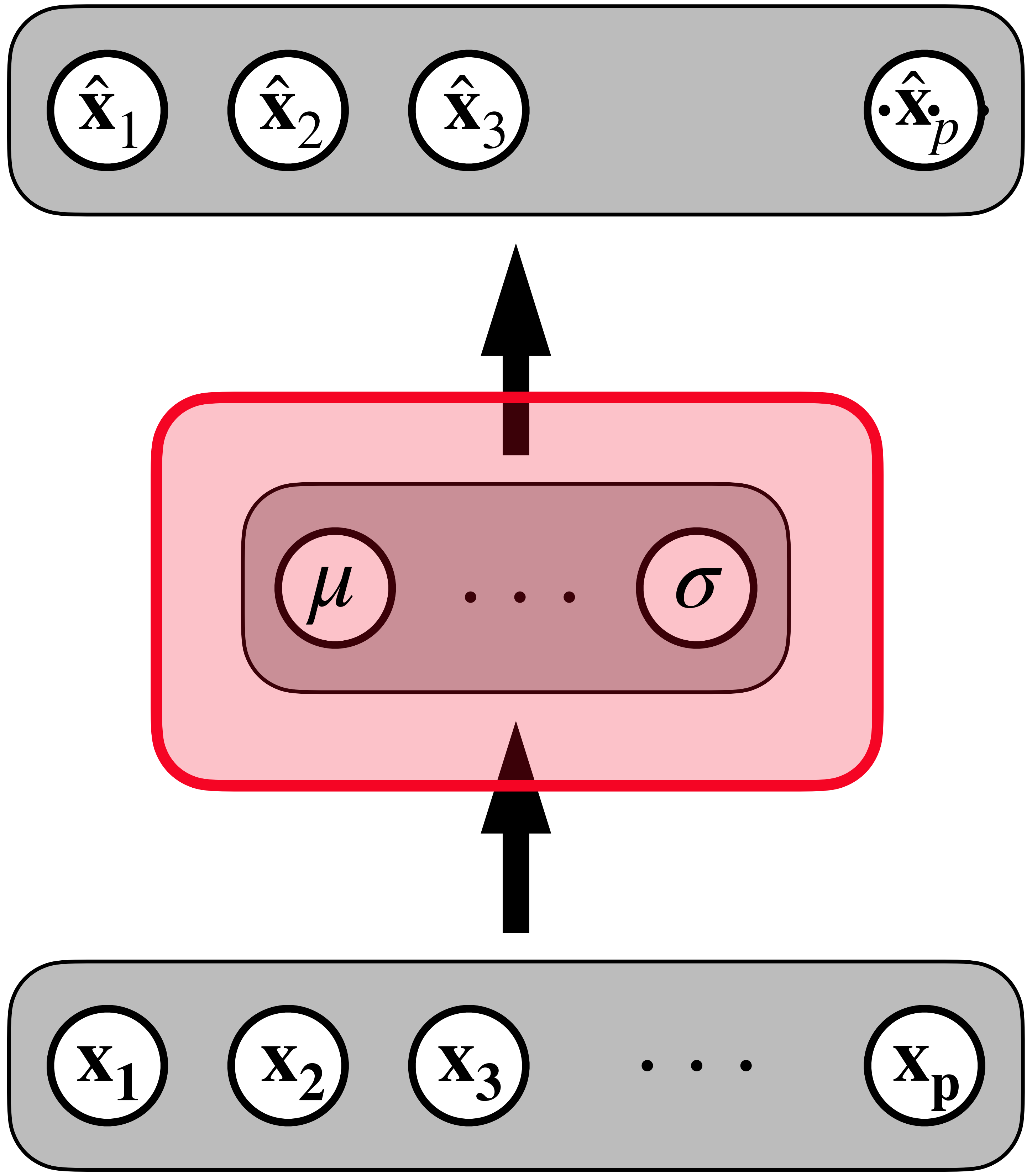

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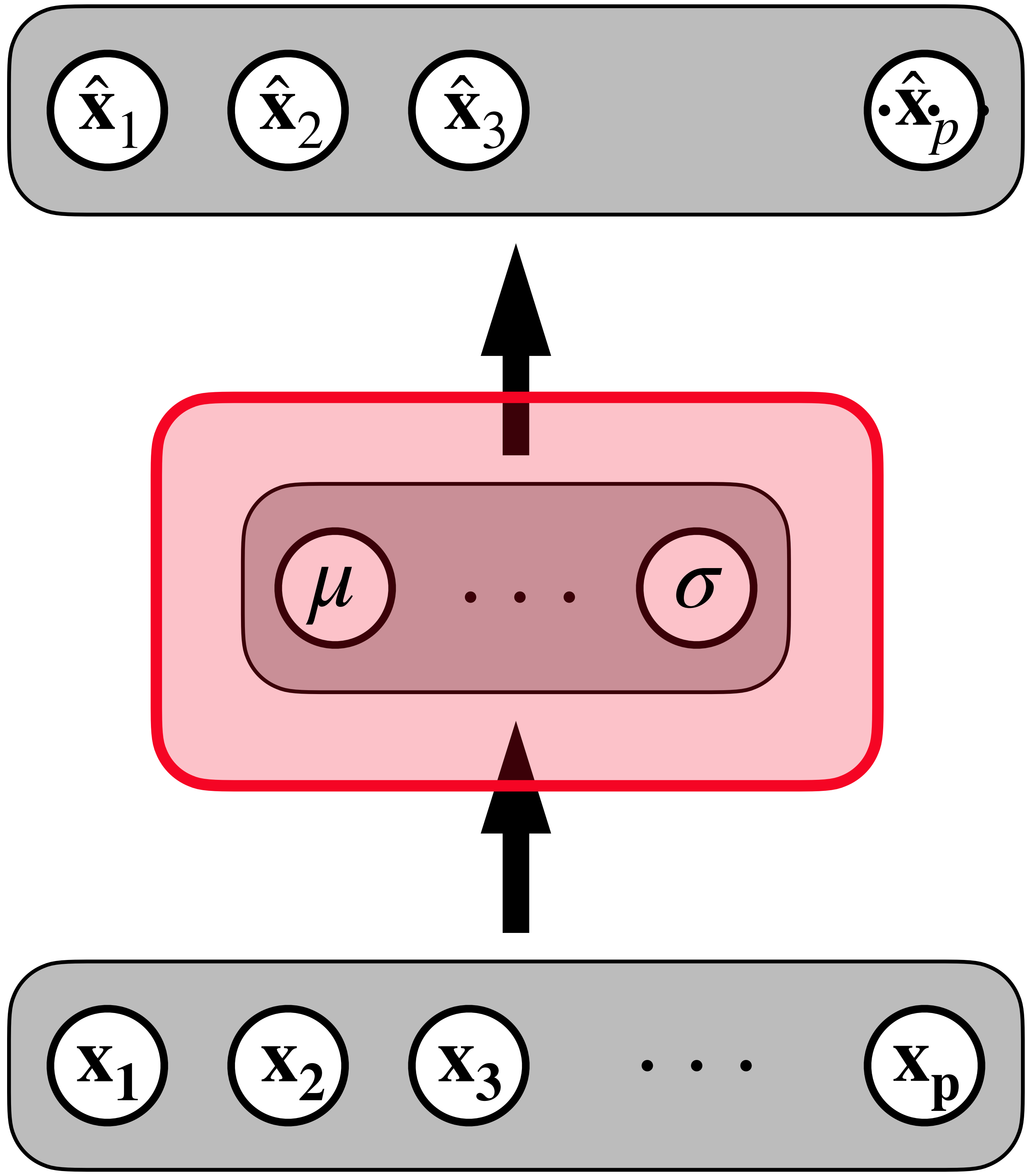

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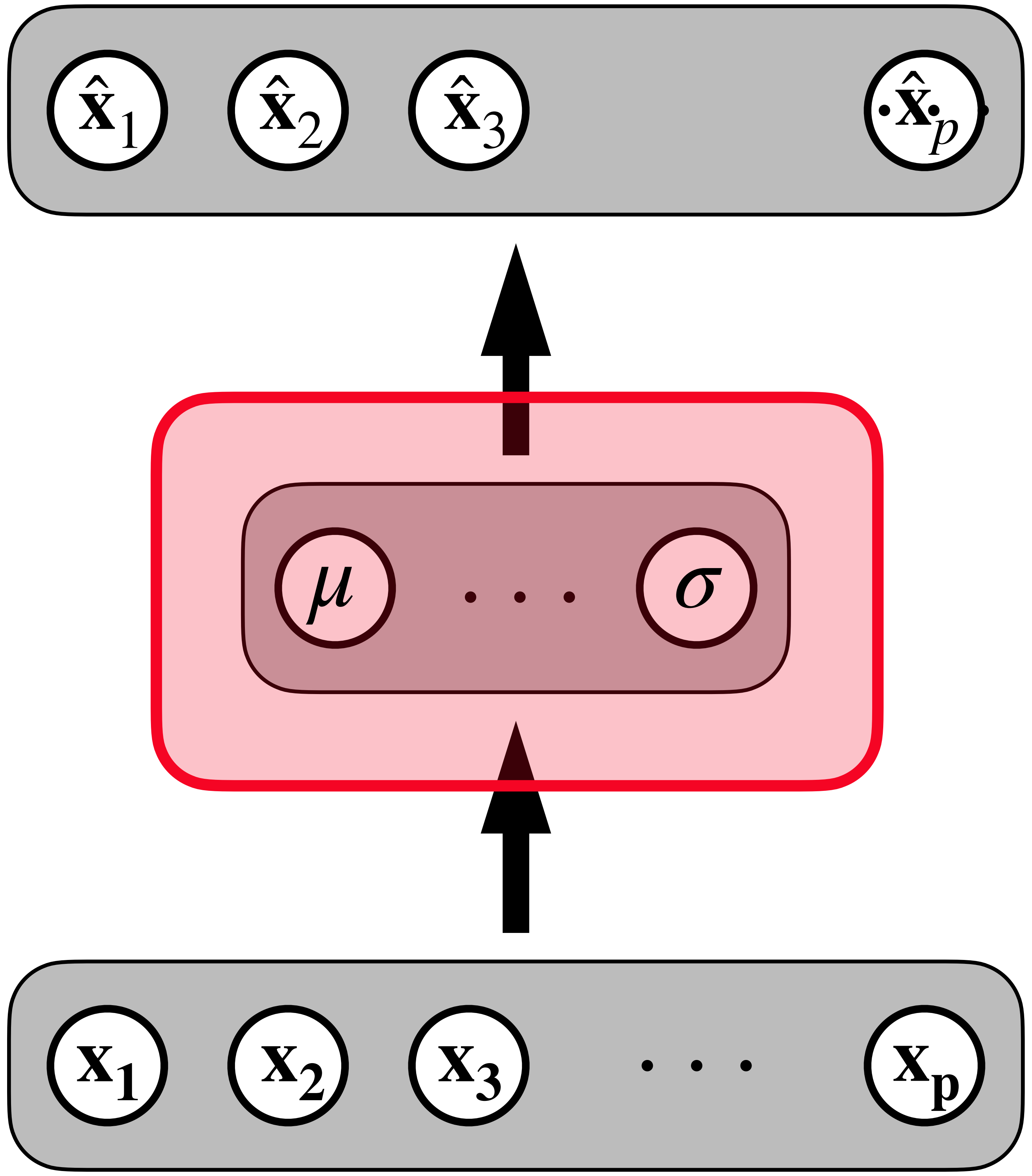

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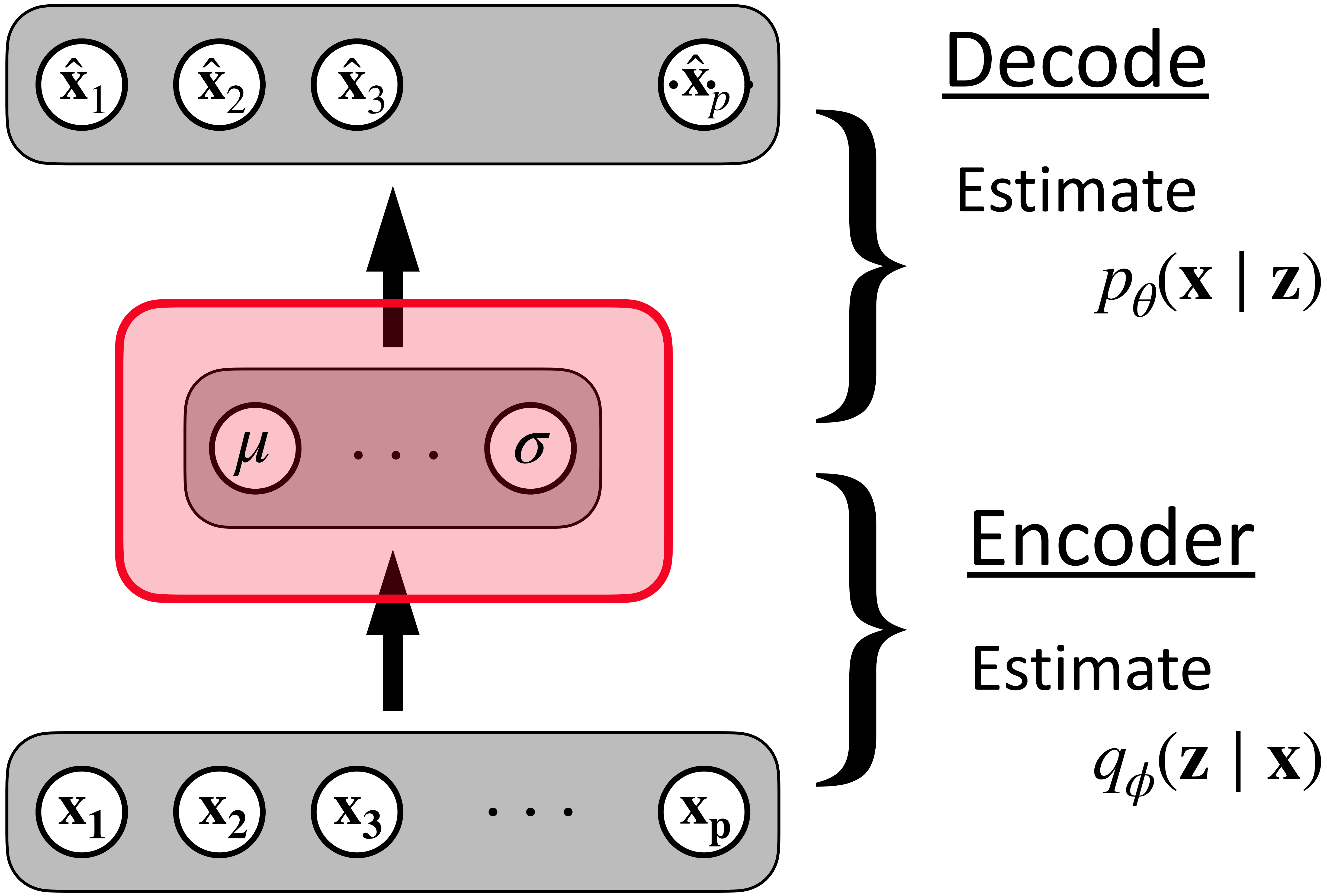

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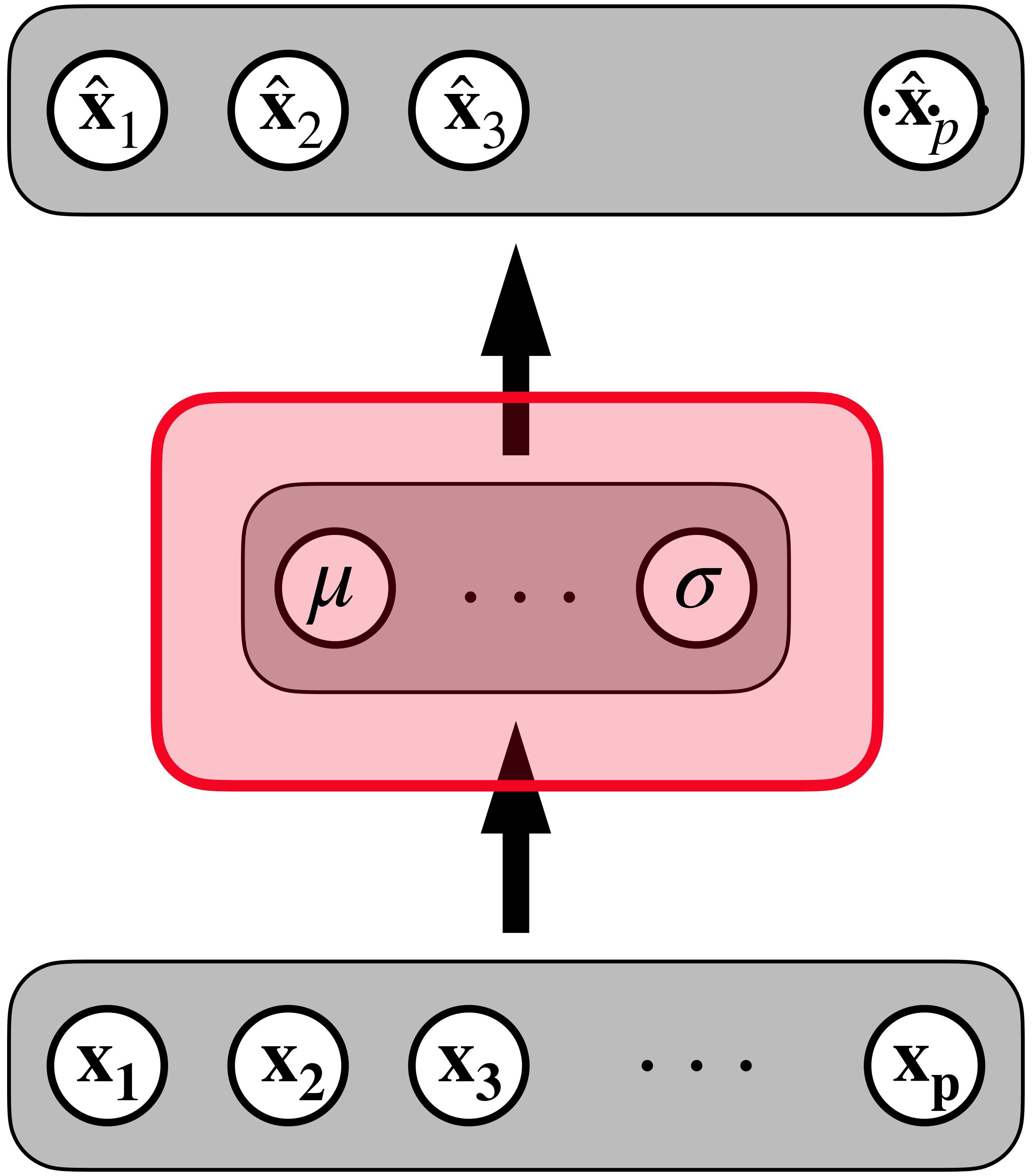
Then we can write $z = \mu + \sigma\epsilon$ où $\epsilon \sim \mathcal{N}(0,1)$.

Variational Auto-encoders

Reparametrize the hidden layer:



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Encoder

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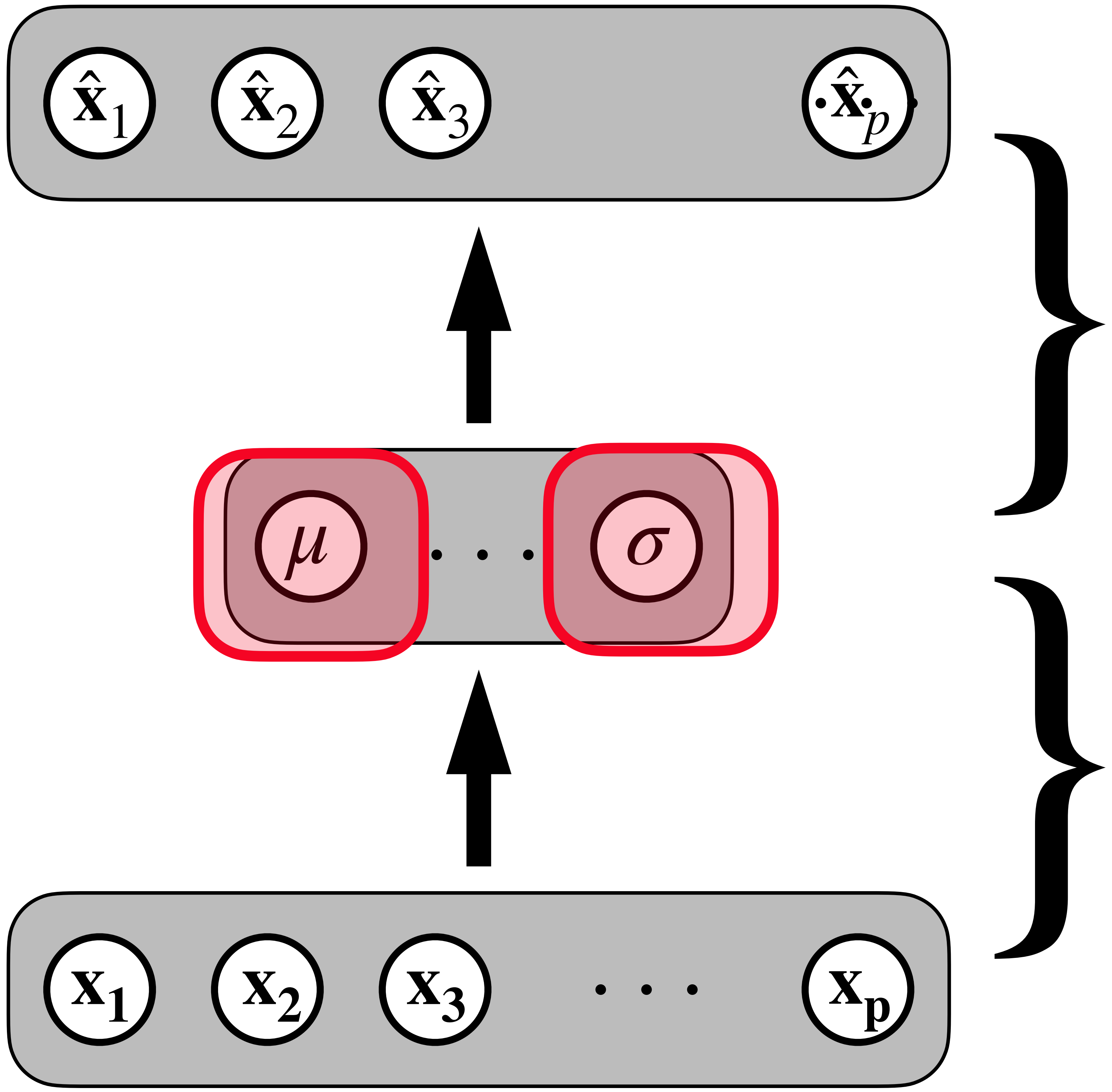

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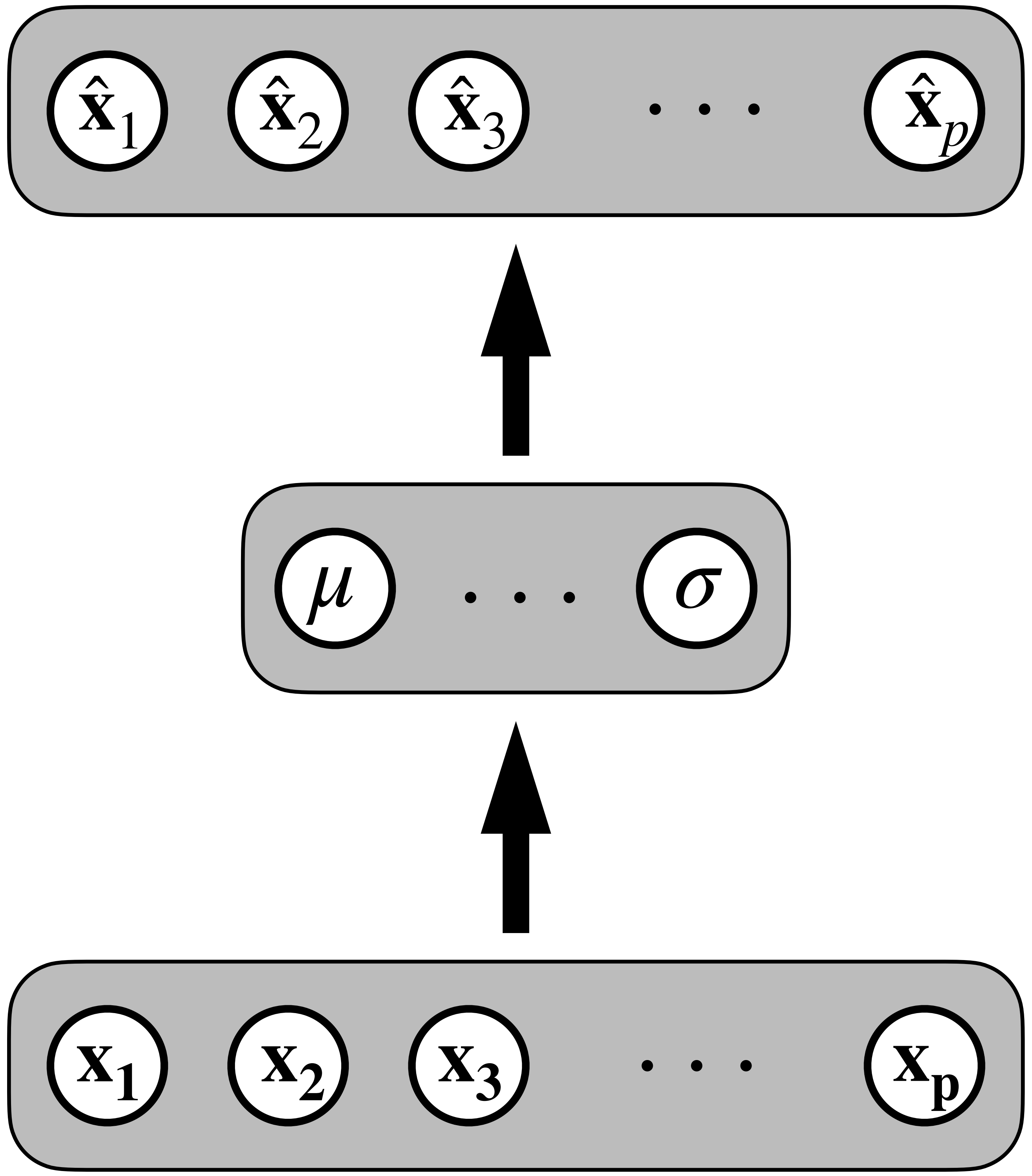

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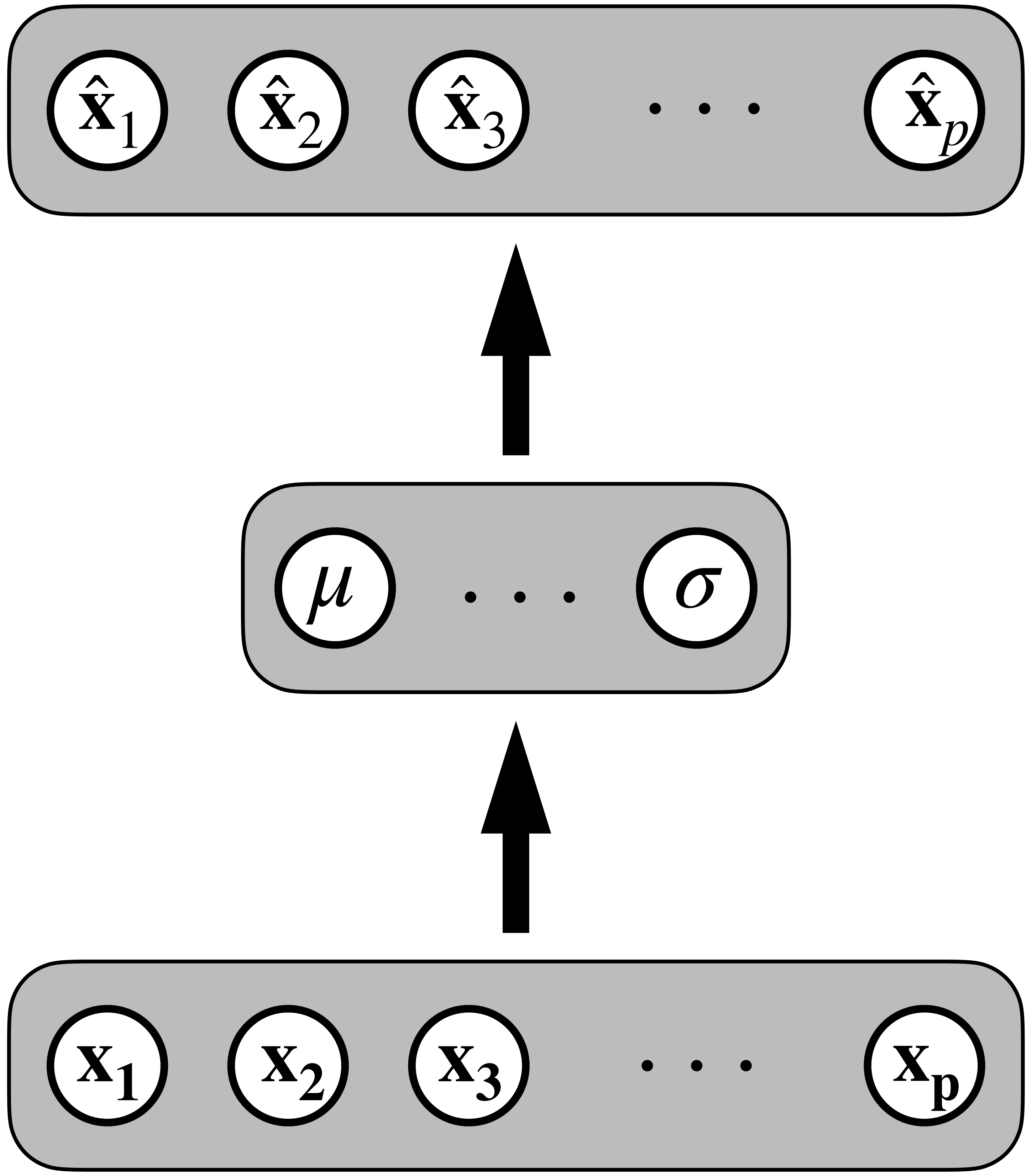
Variational Auto-encoders

VAEs are a way to train a generative model



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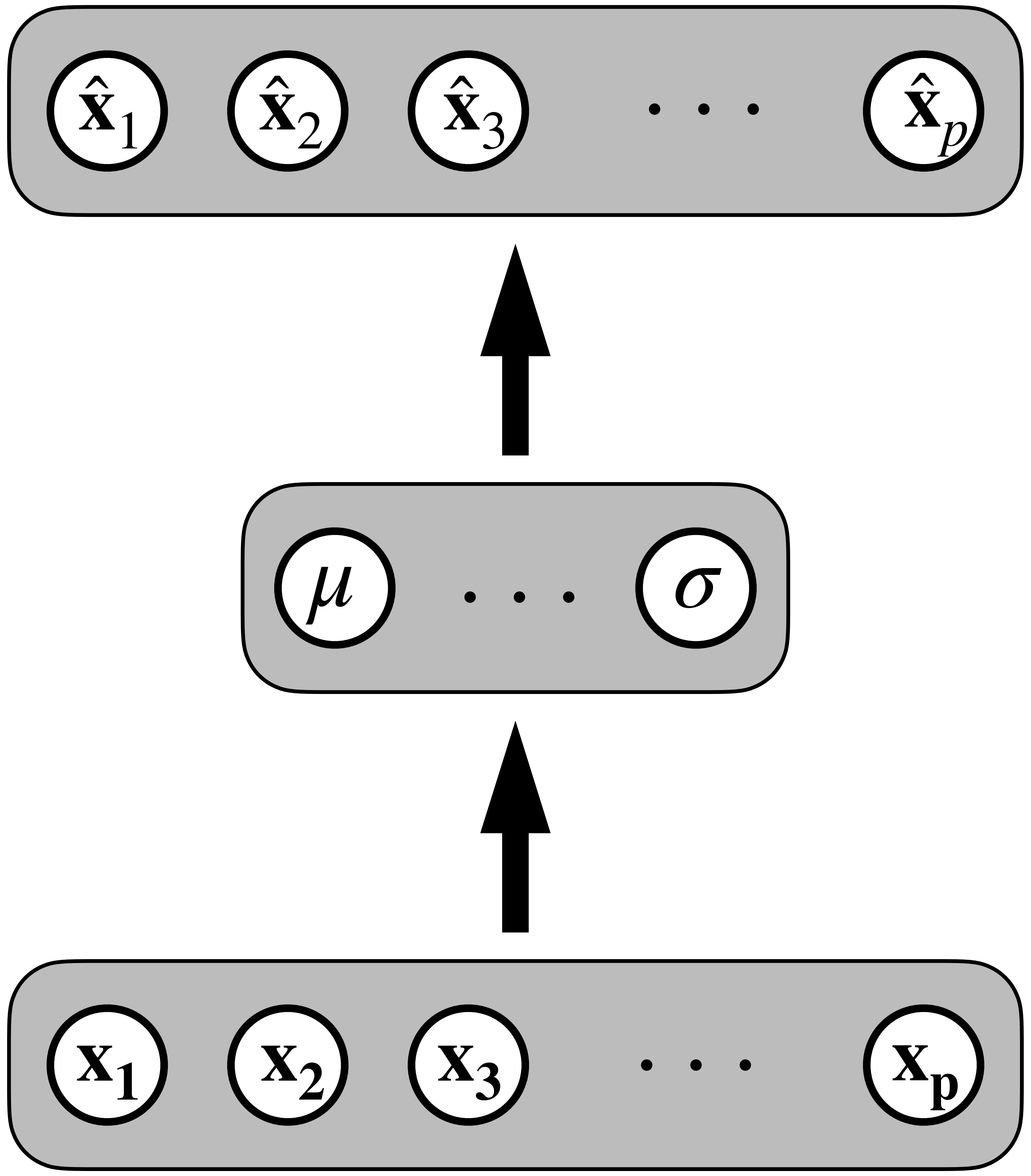

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Variational Auto-encoders

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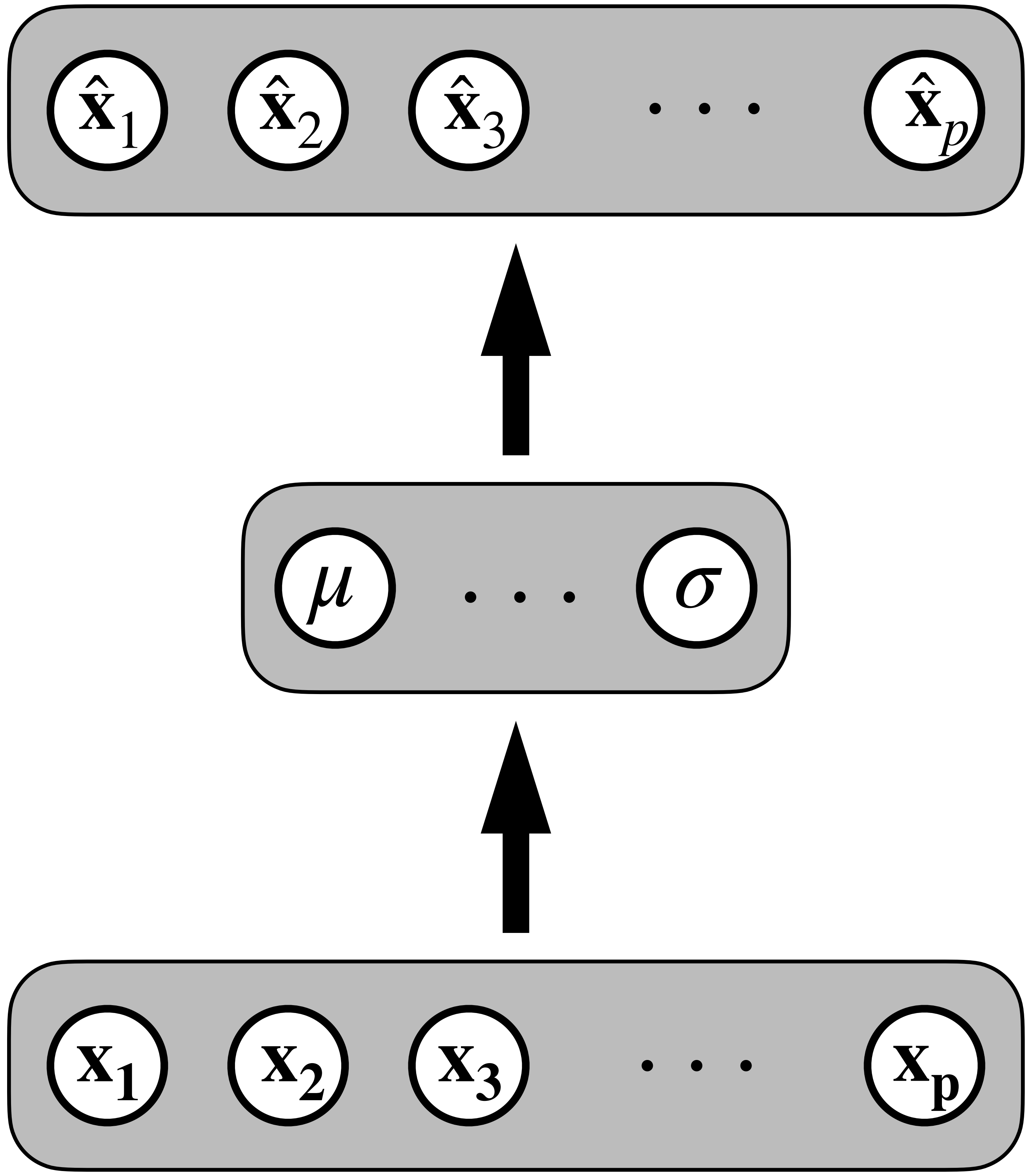

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Variational Auto-encoders

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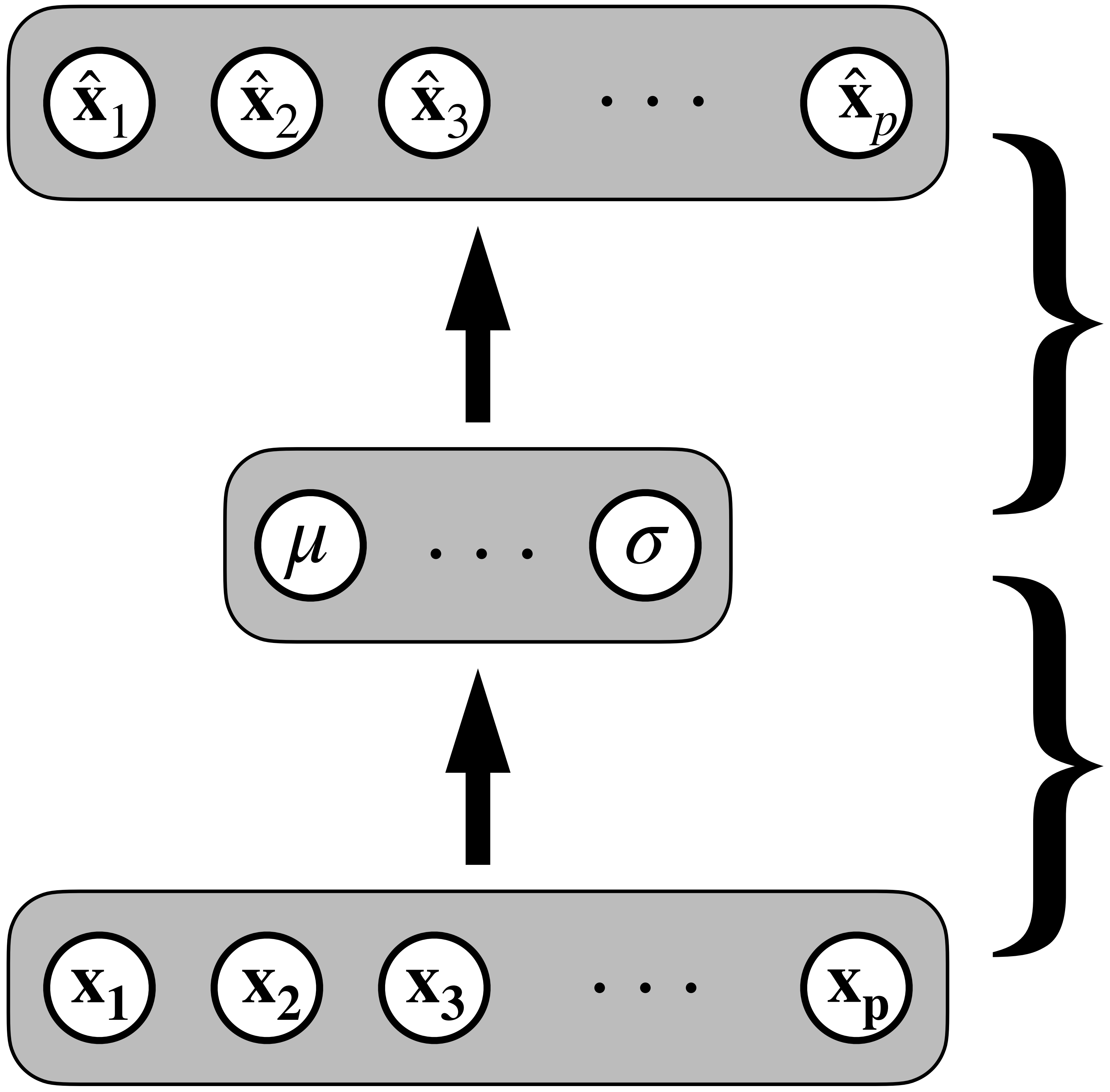

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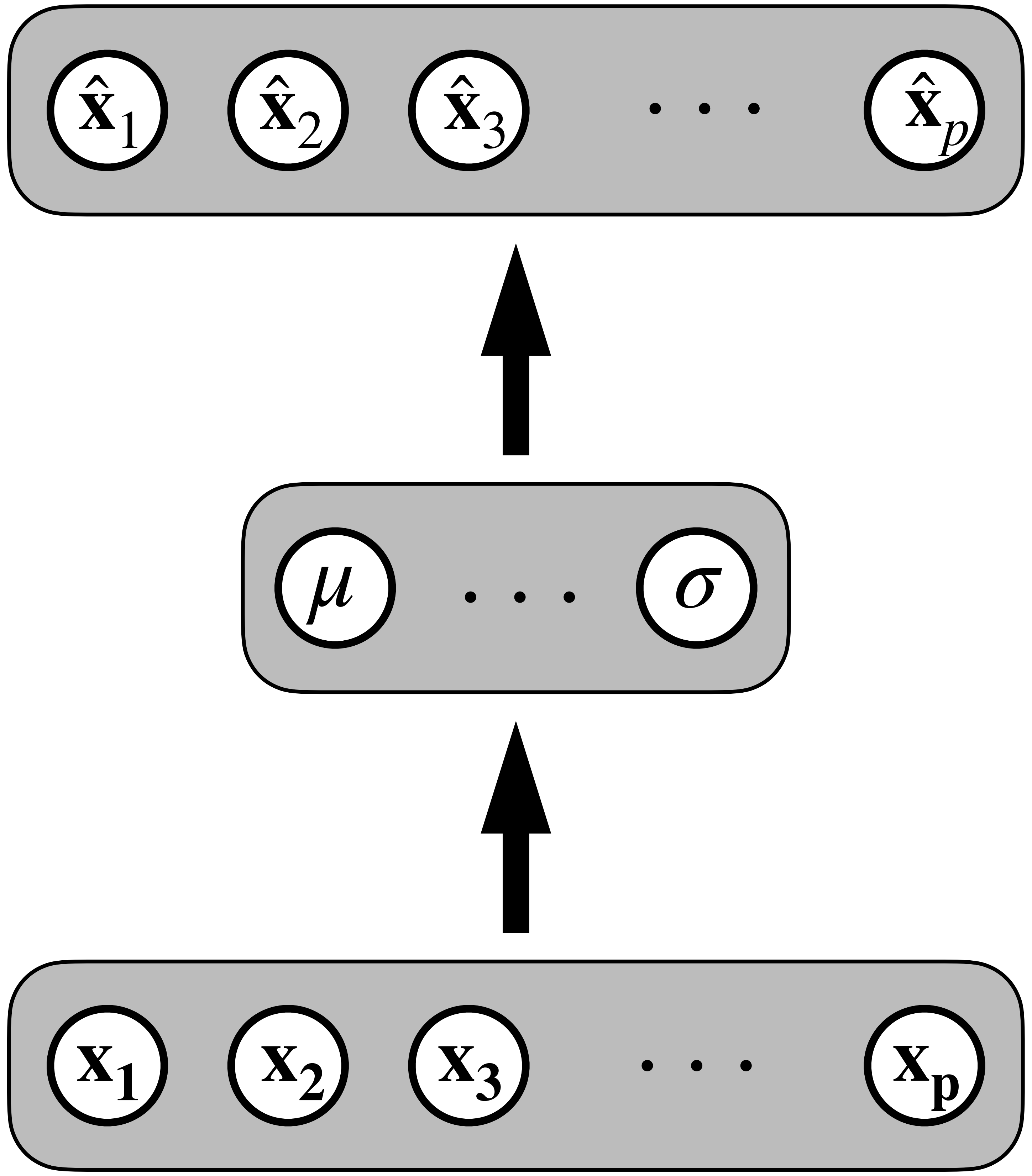

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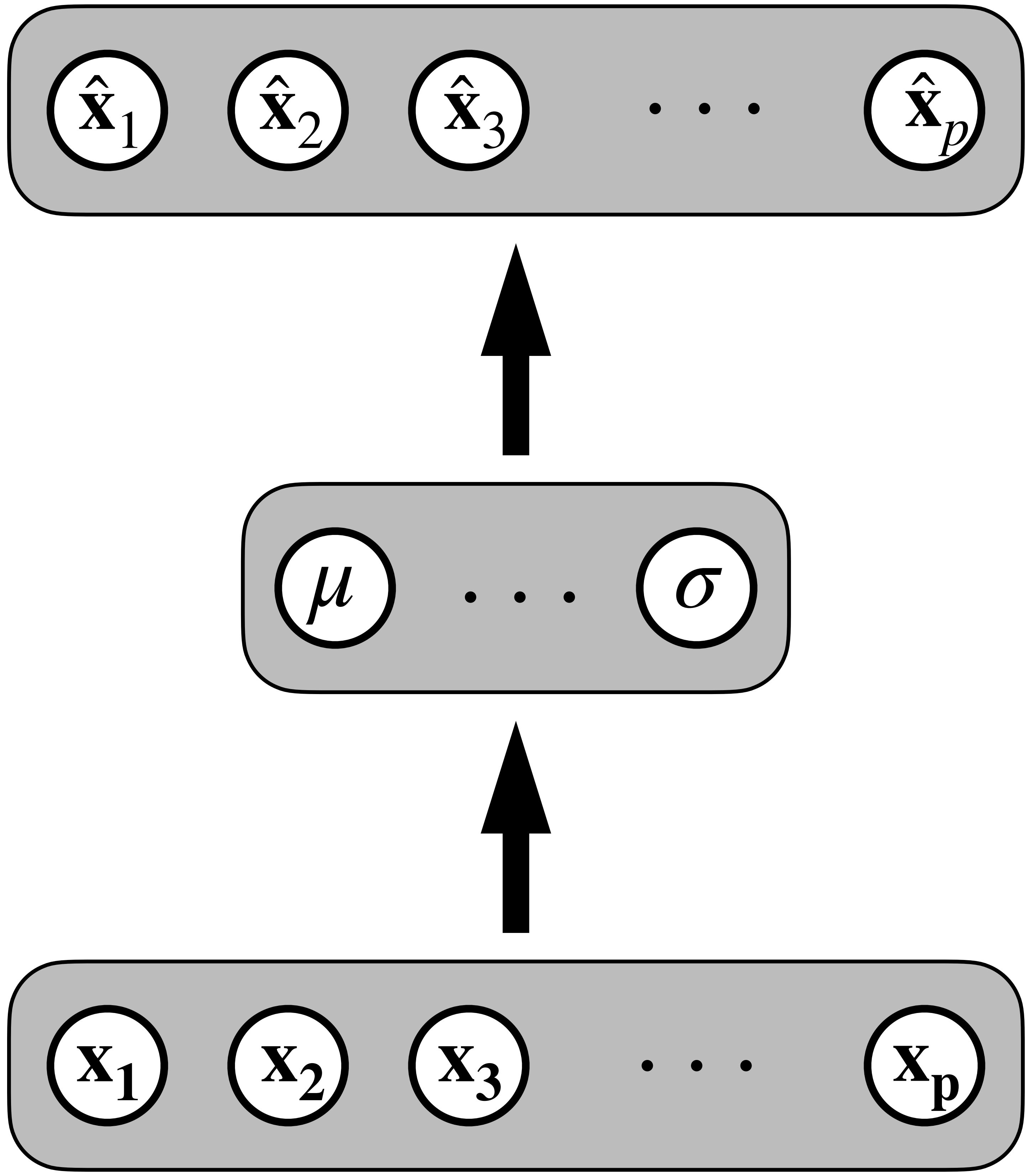

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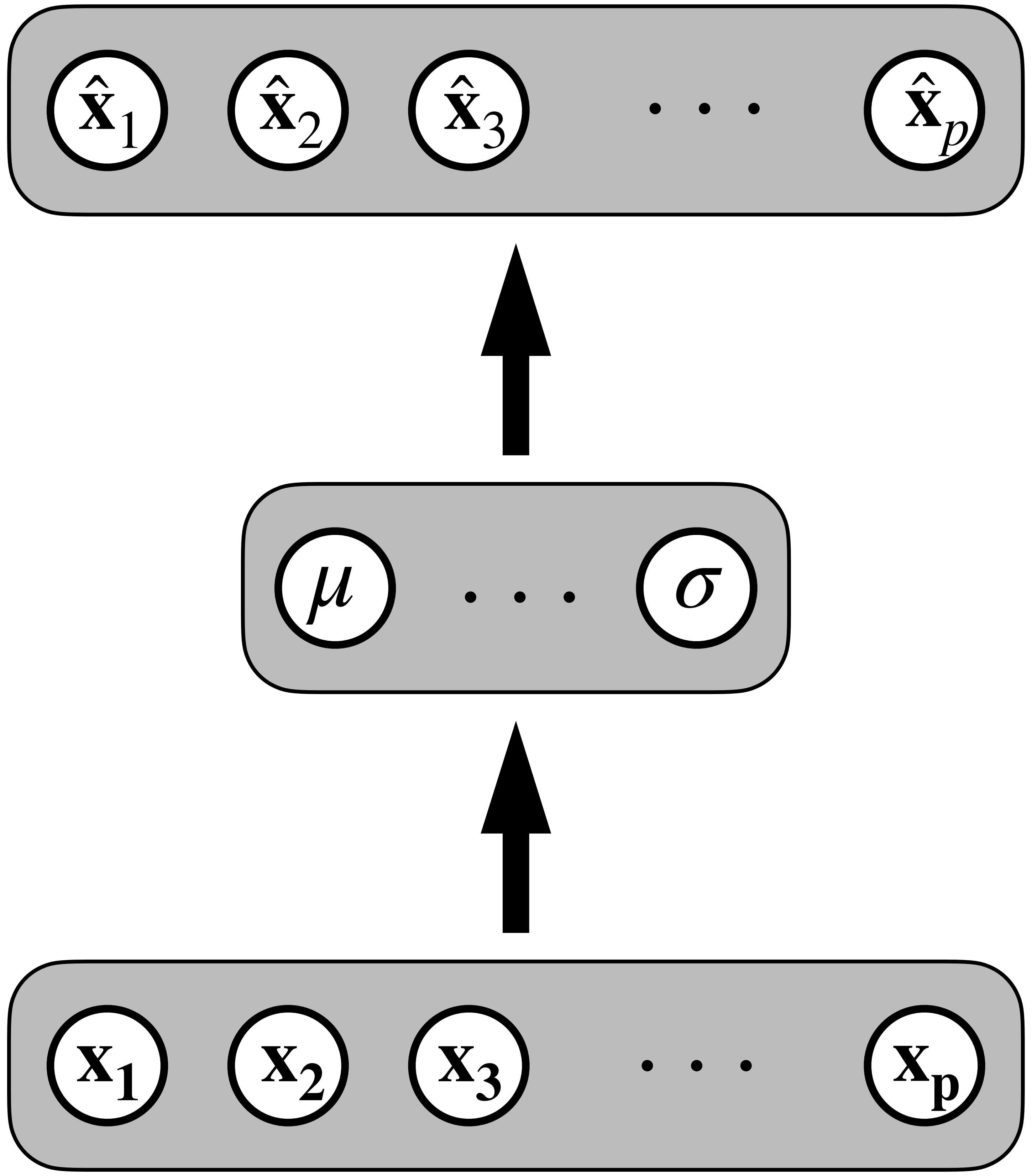

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2. Generate ϵ et obtain $z \sim P(z)$.
3. Obtain $\mathbf{x}$ from $p_\theta(\mathbf{x} \mid \mathbf{z})$.

Variational Auto-encoders

Example: Generating numbers (range 0—9)

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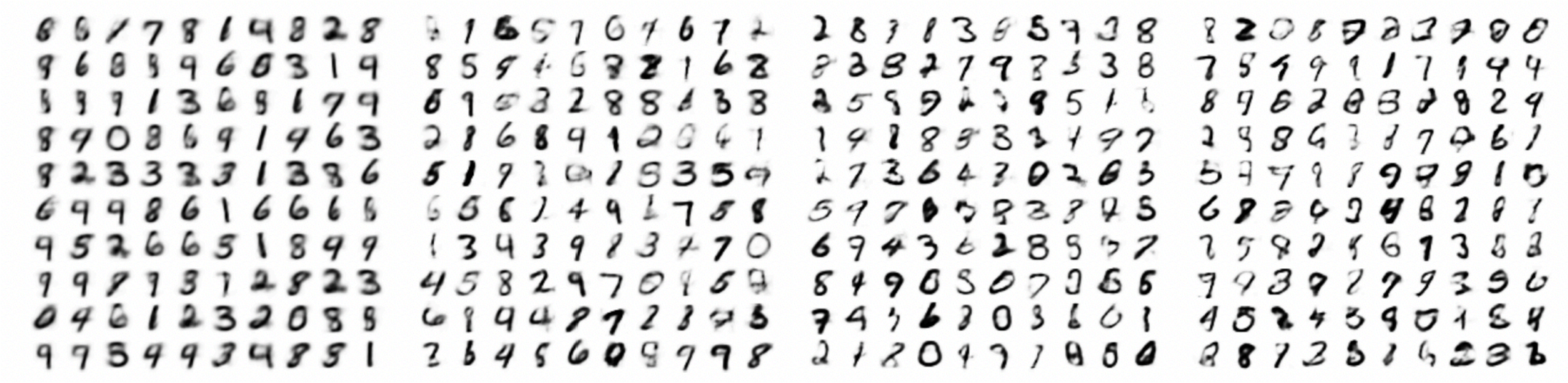


Multivariate Normal ($\mathbb{R}^2$)

Multivariate Normal ($\mathbb{R}^5$)

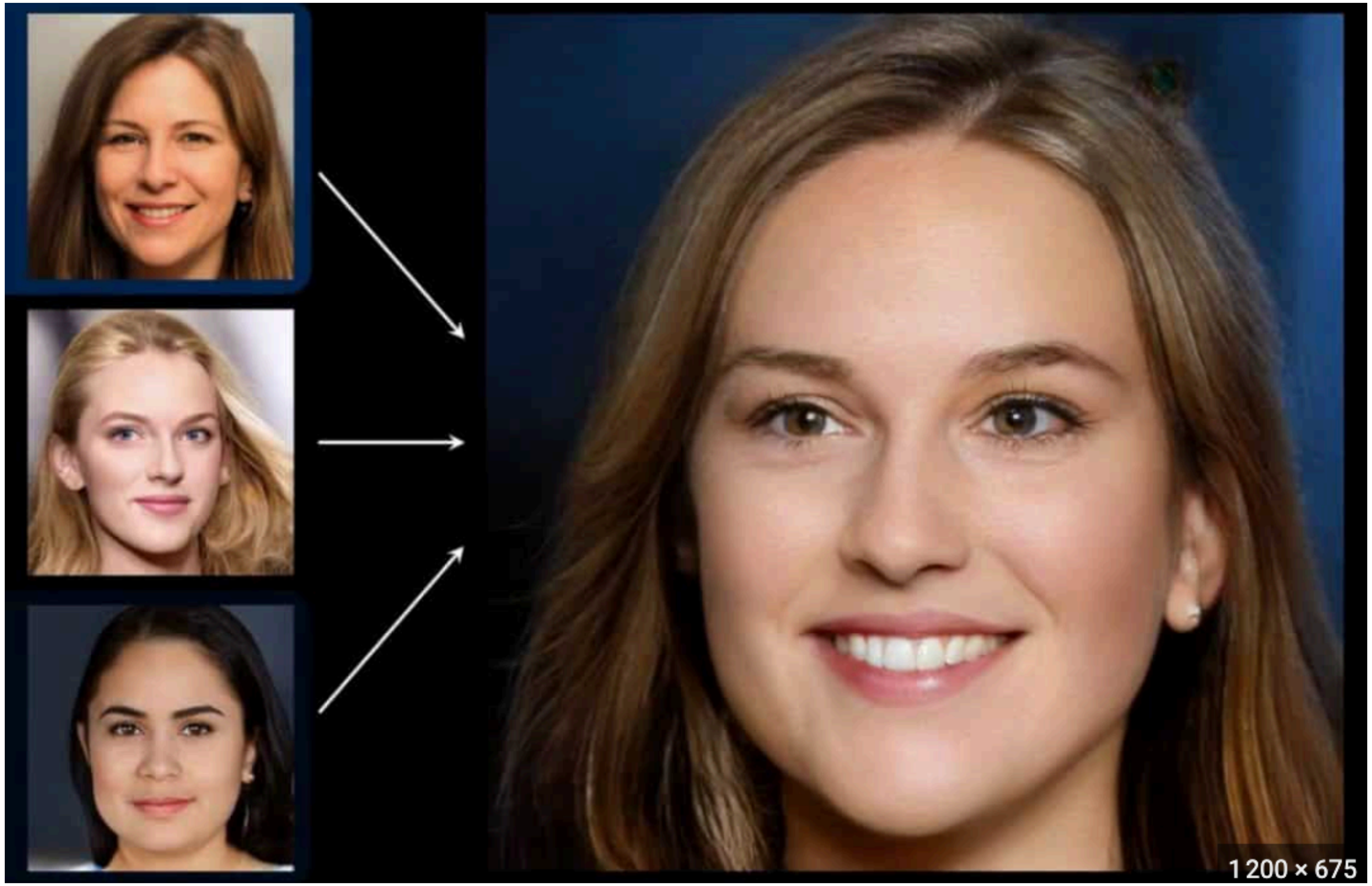
Multivariate Normal ($\mathbb{R}^{10}$)

Multivariate Normal ($\mathbb{R}^{20}$)

Generative Adversarial Networks (GANs)

GANs - Introduction

Well-known for image generation



Historical note

- Framework for learning a generative model (for example, an “inverted” CNN)
- Developed by researchers at Université de Montréal (2014)
- The first to generate high-quality complex images
- Have been (mostly?) replaced by diffusion models
  - The study of GANs has provided insights into 2-player (min-max) optimization

GANs — Intuition

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- Two players (each is a neural network)
  - Generator: The first player. It learns to generate a (good) image
  - Discriminator: The second player, learns to recognize (discriminate) good images from bad images

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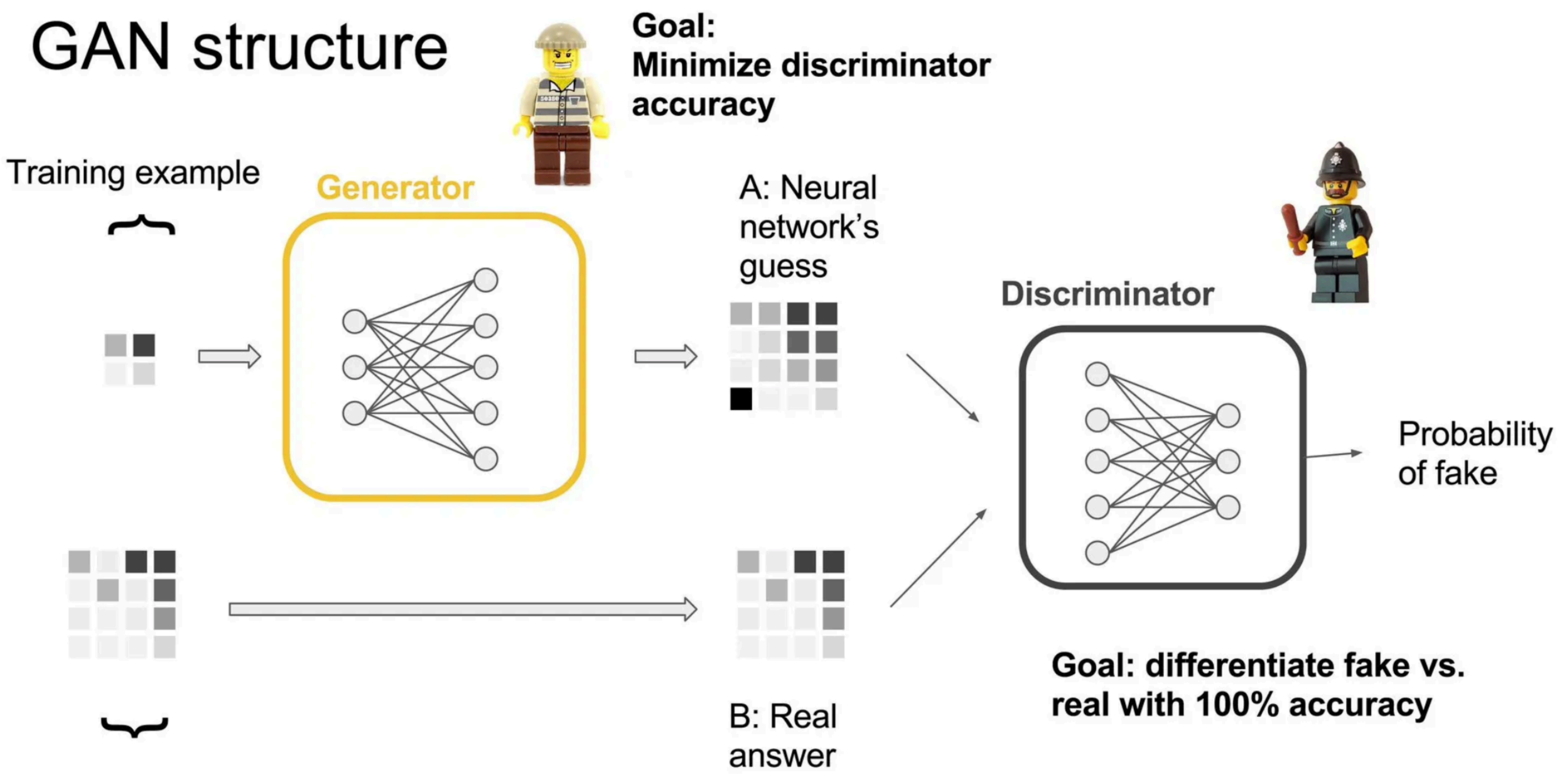
  - Generator: The first player. It learns to generate a (good) image
  - Discriminator: The second player, learns to recognize (discriminate) good images from bad images
- The game (at each round):
  - The second player receives an image. It must determine whether the image comes from the generator or from the training data
    - Depending on the response, the players update (their weights)

GANS — Introduction

Visually:



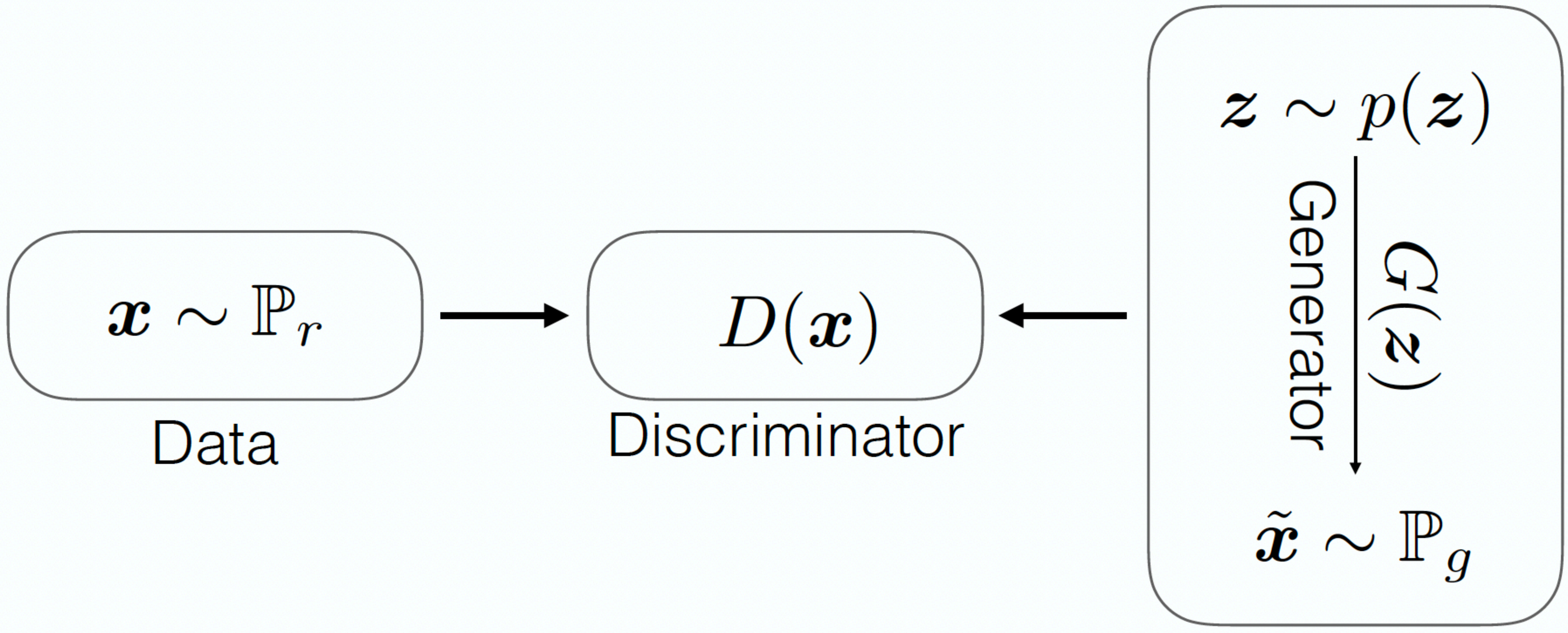
GANs - formalism

$$\boldsymbol{x} \sim \mathbb{P}_r$$

Data

GANs - formalism

GANs - formalism



GANs - objectives

GANs have two objectives (one for each player)

- The output of D is “1” for real and “0" for fake.

Objective of the discriminator

$$\max_{D} \mathop{\mathbb{E}}_{\boldsymbol{x}\sim\mathbb{P}_r} [\log(D(\boldsymbol{x}))] + \mathop{\mathbb{E}}_{\tilde{\boldsymbol{x}}\sim\mathbb{P}_g} [\log(1 - D(\tilde{\boldsymbol{x}}))].$$

Objective of the generator

$$\max_{G} \mathop{\mathbb{E}}_{\tilde{\boldsymbol{x}}\sim\mathbb{P}_g} [\log(D(\tilde{\boldsymbol{x}}))].$$

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GANs - formalism

The game is framed as a two players game between the discriminator and the generator

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Training GANs in practice alternates between training D and G

Algorithm 1 Minibatch stochastic gradient descent training of generative adversarial nets. The number of steps to apply to the discriminator, k, is a hyperparameter. We used $k = 1$, the least expensive option, in our experiments.

---

for number of training iterations **do**

  for k steps **do**

  • Sample minibatch of m noise samples $\{\boldsymbol{z}^{(1)}, \ldots, \boldsymbol{z}^{(m)}\}$ from noise prior $p_g(\boldsymbol{z})$.

  • Sample minibatch of m examples $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ from data generating distribution $p_{\text{data}}(\boldsymbol{x})$.

  • Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^{m} \left[\log D\left(\boldsymbol{x}^{(i)}\right) + \log\left(1 - D\left(G\left(\boldsymbol{z}^{(i)}\right)\right)\right) \right].$$

  end for

  • Sample minibatch of m noise samples $\{\boldsymbol{z}^{(1)}, \ldots, \boldsymbol{z}^{(m)}\}$ from noise prior $p_g(\boldsymbol{z})$.

  • Update the generator by descending its stochastic gradient:

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^{m} \log\left(1 - D\left(G\left(\boldsymbol{z}^{(i)}\right)\right)\right).$$

end for

The gradient-based updates can use any standard gradient-based learning rule. We used momentum in our experiments.

GANs - Varia

Monet ⟲ Photos
Zebras ⟲ Horses
Summer ⟲ Winter
Monet → photo
zebra → horse
summer → winter
photo → Monet
horse → zebra
winter → summer
Photograph
Monet
Van Gogh
Cezanne
Ukiyo-e